INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

SBT-DF204 Computer Forensics Case Studies

Case Study 3: Investigating Memory Evidence

Item	Value
Full name	Nebeuwa Ifeanyichukwu Raphael
Registration number	202520850LE
Course code	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 3: Individual memory forensics investigation
Case title	Investigating Suspected Illegal File Transfer From Memory Evidence
Evidence examined	memdumpWin7.mem (Windows 7 memory image, offline examination)
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1. Executive Summary

Examined the Windows 7 memory image memdumpWin7.mem offline with Volatility 2.6.1 under profile Win7SP1x86_23418 (image time 2019-01-06 15:09:06 UTC), and verified its integrity with a SHA-256 hash (E01). Assessed three questions separately and concluded that evidence of an attempted transfer sequence is moderate, while evidence of a completed transfer and of the person responsible is low.

- **Attempted transfer sequence.** Mapped memory strings referencing http://192.168.56.5/secret_file.docx to browser process chrome.exe, PID 2388 (E03, E13, E16). Identified two closed TCP connections to 192.168.56.5:80 that netscan does not attribute to any process (E04). Recorded the command `copy secret_file.docx F:` entered in cmd.exe (PID 3996; E06, E07) and matched drive F: to a USB mass-storage device (E11, E19). Rated confidence: Moderate.
- **Completed transfer.** Reviewed process and socket records (E03, E04), which show browser activity and connection attempts but give no conclusive confirmation of a finalized download or exfiltration. Noted that the console reported "1 file(s) copied" (E07), but found no independent confirmation: filescan returned no matching file object (E22) and ShellBags contain no F: entry (E23). Rated confidence: Low.
- **Actor attribution.** Linked cmd.exe to the IEUser account (SID ending in -1000; E09, E17) and recorded drive F in IEUser's profile (E19). Established account and device presence only, because the records do not identify the physical person at the keyboard. Considered the sshd_server profile and the sshd.exe service listening on port 22 (E03, E04, E17) as an alternative explanation that the records do not rule out. Rated confidence: Low.

2. Evidence Acquisition and Integrity

Item	Value
Source	Supplied case file for SBT-DF204 Case Study 3. The lab deck (Illegal_File_Transferring_Memory_Forensics.pptx, slide 13) gives the download command, and the downloaded file is the same size as the file listed there, 1,073,676,288 bytes.
Original file name	memdumpWin7.mem
Byte size	1,073,676,288 bytes
SHA-256 (original)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
SHA-256 (working copy)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e (identical)
Original location	~/DF204_CaseStudy3/memdumpWin7.mem; permissions set read-only (chmod 444)
Working copy	~/DF204_CaseStudy3/memdump_working.mem; created with cp; all analysis run on this copy

Item	Value
File times on Kali (ls -l)	Original Oct 19 05:59; working copy Oct 19 06:17
Tools	sha256sum, cp, chmod (Kali Linux); Volatility Foundation Volatility Framework 2.6.1 run as python2 vol.py
Evidence / figures	E01; Fig. 2, Fig. 3

Precaution taken	Why it matters
Hashed the original with SHA-256 before analysis (E01)	The hash is a fingerprint of the file. If a single byte ever changed, the hash would differ, so it can be shown the evidence was not altered after that point.
Set the original read-only (`chmod 444`)	It stops accidental modification or deletion, for example by a mistyped command or a tool that writes to its input.
Made a working copy with `cp` and ran all analysis on it	Any damage or accidental change affects the copy, not the original.
Hashed the working copy and confirmed it matches the original	Identical hashes show the copy is a faithful duplicate, so findings from the copy apply to the original evidence.
Recorded file name, size (1,073,676,288 bytes), file times and source	These details let someone identify the same file and check they hold the same evidence.
Used only read-only Volatility plugins and saved every output with `tee`	Results can be reproduced, and every claim traces back to a recorded command.
Did not contact 192.168.56.5	Examining a memory image should not touch other systems, and connecting could change evidence or go beyond the examiner's authority.

3. Method and Profile Selection

Item	Value
Volatility version	2.6.1, launched as python2 vol.py from ~/DF204_CaseStudy3/volatility
Environment notes	pycryptodome installed so registry plugins load. distorm3 not installed (apihooks, threads, ssdt and mac plugins unavailable; not used). Yara not installed (yarascan unavailable; see Fig. 18).

Item	Value
Candidate profiles	Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
Selected profile	Win7SP1x86_23418
Image date and time	2019-01-06 15:09:06 UTC (local 07:09:06 -0800)
Registry time zone	Pacific Standard Time; Bias 480 minutes = UTC-8 (E05)
Other imageinfo values	KDBG 0x82972c68; DTB 0x185000; PAE; 1 processor

Commands and output files

V = python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418

ID	Plugin	Command (after V)	Output file
E02	imageinfo	python2 vol.py -f ../memdump_working.mem imageinfo	out_imageinfo.txt
E03	pslist	pslist	out_pslist.txt
E04	netscan	netscan	out_netscan.txt
E05	printkey	printkey -K "ControlSet001\Control\TimeZoneInformation"	out_timezone.txt
E06	cmdscan	cmdscan	out_cmdscan.txt
E07	consoles	consoles	out_consoles.txt
E08	hivelist	hivelist	out_hivelist.txt
E09	getsids	getsids -p 3996	out_getsids_cmd.txt
E10	printkey	printkey -K "ControlSet001\Enum\USBSTOR"	out_usbstor.txt
E11	printkey	printkey -K "MountedDevices"	out_mounteddevices.txt
E12	printkey	printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"	out_mountpoints2.txt
E13	strings (GNU)	strings -a -t d ../memdump_working.mem | grep -F "192.168.56.5"	out_strings_ip.txt, ip_strings.txt
E14	printkey	printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList"	out_profilelist.txt
E15	printkey	printkey -K "SAM\Domains\Account\Users\Names"	out_sam_names.txt
E16	strings (Volatility)	strings -s ../ip_strings.txt	out_strings_pid.txt
E17	printkey	printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList\<SID - 1000 / -1002>"	out_profile_1000.txt~, out_profile_1002.txt
E18	printkey	printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}"	out_deviceclasses.txt
E19	printkey	printkey -K "...\Explorer\MountPoints2\F" and "...\MountPoints2\{421e7e52-11dd-11e9-b3cf-08002709e15d}"	out_mp2_F.txt, out_mp2_guid.txt

ID	Plugin	Command (after V)	Output file
E20	printkey	printkey -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0"	out_usbstor_instance.txt
E21	printkey	printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-...}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-...}"	out_dc_usb_link.txt
E22	filescan	filescan | grep -i "secret_file" / grep -i -E "secret|Downloads.*docx"	out_filescan_secret.txt, out_filescan_wide.txt
E23	shellbags	shellbags --output=html --output-file=../shellbags.html	shellbags.html

Profile selection. imageinfo (E02, Fig. 4, App. C1) searched for the kernel debugger block and suggested four Windows 7 x86 profiles: Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000 and Win7SP1x86. It reported KDBG 0x82972c68, DTB 0x185000, PAE paging, Image Type (Service Pack) 1 and one processor. Win7SP1x86_23418 was selected because it was the first suggestion and the service-pack value of 1 fits a Service Pack 1 profile.

Validation. Under this profile pslist returned a coherent process tree: System (PID 4), smss.exe and csrss.exe, with explorer.exe (PID 2524) as parent of chrome.exe, cmd.exe and FTK Imager.exe (E03). netscan returned socket records with owning process IDs and addresses (E04). hivelist resolved the SYSTEM, SOFTWARE, SAM and user hive paths (E08), and printkey read keys from them (E05, E10-E21). getsids and cmdscan/consoles also returned readable results (E09, E06, E07). Readable output from several plugins that depend on the profile structures supports the choice. The evidence log records no run of the other three candidate profiles, so the choice is supported by coherent output, not by a comparison.

Time-zone treatment. All timestamps in the timeline and findings are in UTC, as reported by Volatility. Local time is recorded separately only for reference: the image time of 15:09:06 UTC equals 07:09:06 -0800. The registry confirms Pacific Standard Time, with Bias and ActiveTimeBias of 480 minutes, which is UTC-8 (E05). Local time is therefore UTC minus 8 hours, and no local-time value is used to order events.

4. Findings

Q1. What evidence image did you acquire and how did you preserve it?

Evidence to cite: E01

Observation. The image is memdumpWin7.mem, a Windows 7 memory image of 1,073,676,288 bytes, taken from the supplied case file (E01, Fig. 2). Its SHA-256 value is 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e, and the working copy memdump_working.mem has the identical value (Fig. 3). The original was hashed before analysis, set read-only with chmod 444, and copied with cp; all plugins were run against the working copy. Tools were sha256sum, cp and chmod on Kali Linux and Volatility Framework 2.6.1 (python2 vol.py). Output was saved with tee, and 192.168.56.5 was not contacted.

Interpretation. The matching hashes show the working copy is a faithful duplicate of the file as received, and the read-only setting and copy-first method protect the original from accidental change. Every finding can therefore be traced to a recorded command and a saved output file.

Limit. The hash fixes the file only from the moment it was hashed on the examiner machine. It does not prove the image matches the system at the time of capture, because no hash or chain-of-custody record from the original acquisition is available. The Kali file times (original Oct 19 05:59, working copy Oct 19 06:17) describe handling by the examiner, not the incident.

Q2. What operating-system profile and image time are supported?

Evidence to cite: E02, E05 (validation: E03, E04, E08)

Observation. imageinfo suggested Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000 and Win7SP1x86, and Win7SP1x86_23418 was used (E02). The image date and time is 2019-01-06 15:09:06 UTC (07:09:06 -0800 local). The SYSTEM hive gives Pacific Standard Time with Bias and ActiveTimeBias of 480 (E05). Validation came from coherent pslist, netscan and hivelist output under the chosen profile (E03, E04, E08).

Interpretation. The image is a 32-bit Windows 7 Service Pack 1 system captured at 15:09:06 UTC, and the registry time zone (UTC-8) agrees with the local time that imageinfo reports. The 15:09:06 value is treated as the upper boundary of the timeline: nothing in the image can show events after it.

Limit. The image time is the system clock at capture and cannot be checked against an outside clock. Several other timestamps do not fit it (the 18:02:44-45 start times, the 15:09:07 dllhost.exe start and the 16:31:39 TimeZoneInformation write; see Section 5), so the clock is treated with caution. The other candidate profiles were not run side by side.

Q3. What user and account artefacts are relevant?

Evidence to cite: E08, E09, E14, E15, E17

Observation. hivelist shows hives for two user profiles: IEUser (ntuser.dat 0x8b5d39b0, UsrClass.dat 0xa620b008) and sshd_server (ntuser.dat 0xa65269c8, UsrClass.dat 0xa6539260), plus SYSTEM, SOFTWARE and SAM (E08). The SAM Names key lists Administrator, Guest, IEUser, sshd and sshd_server (E15). ProfileList has S-1-5-18, -19, -20 and the SIDs ending -1000 and -1002 (E14). SID -1000 maps to C:\Users\IEUser and its key was last written 2019-01-06 15:07:15 UTC; SID -1002 maps to C:\Users\sshd_server and was last written 2015-09-21 09:51:49 UTC (E17). The token of cmd.exe (PID 3996) belongs to IEUser (S-1-5-21-1716914095-909560446-1177810406-1000) and includes Administrators, Interactive and Console Logon (E09).

Interpretation. The command shell that typed the copy command ran under the IEUser account, with administrator rights, in an interactive console session. The IEUser profile is the one that holds the Downloads folder bag and the F: drive mapping (E19, E23). The sshd_server profile has an old last-write date and no activity tied to the events in this report.

Limit. Profile and token records identify an account, not the person at the keyboard. The IEUser password could be shared, the session could have been used by someone else, and sshd.exe was listening on port 22 (E03, E04), so remote access through the sshd_server service account cannot be excluded from this evidence alone. The -1000 key write at 15:07:15 matches the FTK Imager.exe start to the second (E03); the cause was not determined.

Q4. What process and network evidence relates to a potential download or transfer?

Evidence to cite: E03, E04, E13, E16

Observation. chrome.exe PID 2388 (parent explorer.exe PID 2524) started at 15:04:30, with child chrome.exe processes 1280 and 2380 (15:04:30), 2912 (15:04:32) and 976 (15:06:23) (E03). netscan shows two TCPv4 records from 192.168.56.8 (ports 49177 and 49178) to 192.168.56.5:80, both CLOSED, PID -1, with no owner or creation time (offsets 0x3e043b10 and 0x3fcba008). It also shows a chrome.exe (PID 2388) UDPv4 socket 192.168.56.8:56647 created at 15:08:32 and sshd.exe (PID 2016) listening on port 22 (E04). Strings for 192.168.56.5 mapped to chrome.exe PID 2388 include http://192.168.56.5/secret_file.docx (offset 152263320, [2388:072eda98]), a cached header at offset 568756597 ([2388:0720c975]) with Last-Modified Fri, 04 Jan 2019 16:05:12 GMT and a Word-document content type, and a Google omnibox request for 192.168.56.5/secret_file.docx (offset 799213234). Other hits sit in free memory (E13, E16).

Interpretation. Chrome memory holds the address of the file and a response header for a Word document from that server, so Chrome handled that URL after it started at 15:04:30. The closed connections to port 80 of the same server are consistent with Chrome requests, but they have no owning process, so the link between them and Chrome is an inference. The Last-Modified value is the server time of the file (4 Jan 2019, GMT = UTC), not a download time. The Google search strings show the text was also entered as search text, and that on its own does not contact 192.168.56.5.

Limit. Strings in memory show that data was handled, not that the file finished downloading, was saved or opened. Many hits are in free memory with no owner. The header record also contains the variant 192.128.56.5, which was not explained. No endpoint was contacted; all work was done offline on the image.

Q5. What does command history show?

Evidence to cite: E06, E07

Observation. cmdscan shows conhost.exe PID 3988 with a cmd.exe command history of three commands: Cmd #0 ipconfig, Cmd #1 cd Downloads, Cmd #2 copy secret_file.docx F: (E06). conhost.exe PID 2000 (sshd.exe) has no commands. The consoles output for cmd.exe PID 3996 shows the prompt C:\Users\IEUser, ipconfig reporting IPv4 192.168.56.8, then C:\Users\IEUser\Downloads>copy secret_file.docx F: followed by "1 file(s) copied." (E07). cmd.exe PID 3996 and conhost.exe PID 3988 started at 15:06:11 (E03).

Interpretation. The commands were typed between 15:06:11 and the capture at 15:09:06, after drive F: was mapped (15:03:06-07) and after Chrome started (15:04:30). The commands used a relative file name inside the IEUser Downloads folder, so a file named secret_file.docx was probably in that folder when the command ran, and the console text reports that one file was copied to F:. The IPv4 address matches the local address in netscan (E04).

Limit. There is no per-command timestamp. The console line is screen text, not independent proof that the file reached the USB device with its content intact; filescan found no file object (E22). The file name matches the URL, but the memory does not prove it is the same file that Chrome requested.

Q6. What removable-media evidence is present?

Evidence to cite: E10, E11, E12, E18, E19, E20, E21

Observation. The USBSTOR key has one device, Disk&Ven_General&Prod_UDisk&Rev_5.00, instance 6&1bec0f48&0&_&0, with FriendlyName "General UDisk USB Device" (E10, E20). The USBSTOR, instance, DeviceClasses and MountedDevices keys were all last written at 15:02:54 UTC (E10, E11, E18, E20, E21). MountedDevices maps \DosDevices\F: and the volume {421e7e52-11dd-11e9-b3cf-08002709e15d} to the same USBSTOR string, and \DosDevices\E: is a VirtualBox CD-ROM (E11). In the IEUser ntuser.dat, MountPoints2\F was last written at 15:03:06 and the volume GUID key at 15:03:07 (E12, E19). The second character of the instance ID is "&" (E20).

Interpretation. A USB mass-storage device was present by 15:02:54, at about the time explorer.exe started, and was assigned drive letter F: and registered in IEUser's profile seconds later. That links F: in the copy command to this device and to the IEUser account, before Chrome and cmd.exe started.

Limit. These keys show device presence and mounting, not file writes. A last-write time shows the most recent update, not the first connection or removal. The instance ID has no serial number, so the device cannot be told apart from another of the same model, and ownership of the device is not shown.

Q7. What File Explorer evidence supports the sequence?

Evidence to cite: E23 (and E22)

Observation. The IEUser ShellBags (UsrClass.dat, BagMRU\0\1\1\0) contain C:\Users\IEUser\Downloads with key LastWrite 2019-01-06 15:04:40 UTC and an Access field of 2019-01-06 14:49:40 UTC (E23). D:\ and E:\ entries were last written in 2015, and there is no F:\ entry. The Desktop bag LastWrite is 14:32:53 UTC. Two filescan searches for secret_file and for Downloads.*docx returned no output (E22).

Interpretation. The Downloads folder bag was updated at 15:04:40, ten seconds after Chrome started and before cmd.exe started at 15:06:11, which is consistent with the folder being viewed before the command session. The absence of F:\ does not contradict the copy, because a command-line copy does not use File Explorer and so would not create a ShellBag.

Limit. The LastWrite is the time the registry key changed, and the Access field is the folder's own timestamp stored in the shell item; neither is a creation or modification time of secret_file.docx, and neither shows that the file was in the folder. The 14:32:53 and 14:49:40 values are earlier than this boot session (see Section 5). A negative filescan result shows only that no file object was found in memory; it does not show the file was absent.

Q8. What conclusion can you defend?

Evidence to cite: Timeline (Section 5) and Attribution assessment (Section 6)

Conclusion. The memory evidence supports a cautious conclusion: the IEUser account, in an interactive session, had a Chrome process holding the address http://192.168.56.5/secret_file.docx and a Word-document response header, had a USB device mapped as F:, and ran cd Downloads and copy secret_file.docx F: in cmd.exe, which printed "1 file(s) copied.", all before the capture at 2019-01-06 15:09:06 UTC. This is evidence of an attempted transfer sequence. It does not on its own prove that the file was downloaded intact, that it reached the USB device, or who was at the keyboard.

Confidence. Attempted transfer sequence: Moderate. Completed transfer: Low, because filescan and ShellBags give no independent confirmation (E22, E23). Actor attribution: Low (Section 6).

Limitations and alternative explanations. (1) The image is a single snapshot with no per-command timestamps, so the order of the commands against the Chrome and network activity is inferred from process start times. (2) The two TCP connections to 192.168.56.5 are CLOSED with no owner, and many URL strings sit in free memory, so some of them could come from a search or typed text and not a completed request. (3) The sshd.exe service on port 22 and the sshd_server profile mean another way of access cannot be ruled out. (4) Several timestamps are inconsistent with the image time (Section 5), so the clock is treated with caution. (5) The USB instance has no serial number, so the device cannot be uniquely identified. (6) FTK Imager.exe was running from 15:07:15, so the live system was changing while it was captured.

5. Reconstructed Timeline (UTC)

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Server-side file time (cache header)	2019-01-04 16:05:12	E13, E16	Chrome PID 2388 memory; string at file offset 568756597 ([2388:0720c975])	Cached HTTP header for http://192.168.56.5/secret_file.docx shows Last-Modified Fri, 04 Jan 2019 16:05:12 GMT and a Word-document content type.	Last-Modified is the web server's time for the file (about two days before capture; GMT equals UTC). It shows when the file last changed on the server, not when it was downloaded or opened here. Limit: it comes from one header fragment in Chrome memory.
Folder-view evidence (before this boot session)	2019-01-06 14:49:40	E23	ShellBags, UsrClass.dat, BagMRU\0\1\1\0	Shell item for C:\Users\IEUser\Downloads records Access 14:49:40 (earlier than the first processes of this session; see inconsistencies below).	The Access value stored in the Downloads shell item is earlier than the first processes of this session, so it records an earlier moment than this boot. It is the folder's own timestamp, not a record of the file or the copy. Used as background only (see inconsistencies).
System / session context	2019-01-06 15:02:54	E03	pslist	explorer.exe PID 2524 starts. (Earliest processes are listed at 18:02:44-45; see inconsistencies below.)	Start of IEUser's interactive shell. chrome.exe, cmd.exe and FTK Imager.exe are its children, so it anchors user activity in this session. Limit: the earliest system processes (18:02:44-45) conflict with it (see inconsistencies).
Removable-device evidence	2019-01-06 15:02:54	E10, E11, E18, E20, E21	SYSTEM hive: USBSTOR, MountedDevices, DeviceClasses keys	USBSTOR instance, DeviceClasses interface link and MountedDevices keys are all last written at 15:02:54. FriendlyName: General UDisk USB Device.	A USB mass-storage device was present by this time, before Chrome and cmd.exe. A last-write time shows the latest update, not the first plug-in. The instance ID has no serial number, so the device cannot be uniquely identified.
Drive mapping	2019-01-06 15:03:06 - 15:03:07	E11, E12, E19	IEUser ntuser.dat MountPoints2; SYSTEM MountedDevices	MountPoints2\F last written 15:03:06; volume GUID {421e7e52-11dd-11e9-b3cf-08002709e15d} key last written 15:03:07. MountedDevices maps \DosDevices\F: to the USBSTOR device.	Drive F: was assigned to the USB device and registered in IEUser's profile within about 12 seconds of the device keys. This ties F: to the device and to IEUser. It shows the drive was mounted, not that any file was written to it.
Browser or process evidence	2019-01-06 15:04:30	E03	pslist	chrome.exe PID 2388 (parent explorer.exe 2524) starts; child chrome.exe PIDs 1280 and 2380 (15:04:30), 2912 (15:04:32), 976 (15:06:23).	Chrome started about 1.5 minutes after the drive was mapped. Strings mapped to PID 2388 were handled by this process after it started (inference). Limit: the strings have no timestamps and do not show a completed download.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Download-folder review	2019-01-06 15:04:40	E23	ShellBags, UsrClass.dat, BagMRU\0\1\1\0	LastWrite of the key for C:\Users\IEUser\Downloads (sibling entries share the same LastWrite).	Written 10 seconds after Chrome started and before cmd.exe started, consistent with the Downloads folder being viewed before the commands. It is a registry key write time, not a file creation or modification time, and does not show secret_file.docx was there.
Command-history evidence (process start)	2019-01-06 15:06:11	E03	pslist	cmd.exe PID 3996 (parent explorer.exe 2524) and conhost.exe PID 3988 start.	cmd.exe started about 1 minute 41 seconds after Chrome and about 3 minutes before capture. This is the earliest possible time for the typed commands. Limit: it is a process start time, not a command time.
Command-history evidence (commands)	Between 15:06:11 and 15:09:06 (no per-command time)	E06, E07	cmdscan; consoles (conhost.exe PID 3988)	In order: ipconfig; cd Downloads; copy secret_file.docx F: . The console buffer then shows "1 file(s) copied."	The commands appear in order and fall between 15:06:11 and 15:09:06, after the drive mapping and Chrome start. The ipconfig address 192.168.56.8 matches netscan. Limit: no per-command times, and the console line is text, not independent proof the file reached F:.
Tool activity before capture	2019-01-06 15:07:15	E03	pslist	FTK Imager.exe PID 2596 starts (parent explorer.exe 2524).	FTK Imager.exe was running before capture, consistent with the imaging being under way (inference). Imaging a live system changes it, so the state at 15:09:06 may include the tool's effect. The IEUser profile key was also written at 15:07:15; the cause was not determined.
Browser network artefact	2019-01-06 15:08:32	E04	netscan	chrome.exe PID 2388 UDPv4 socket 192.168.56.8:56647 created. (The two TCP records to 192.168.56.5:80, offsets 0x3e043b10 and 0x3fcba008, are CLOSED, PID -1, no creation time.)	A Chrome UDP socket created about 34 seconds before capture shows network activity near the end. It is not an HTTP session to 192.168.56.5. The two TCP records to that server are CLOSED with no PID or time, so they cannot be placed on this timeline or tied to Chrome.
Image capture	2019-01-06 15:09:06	E02	imageinfo	Image date and time 2019-01-06 15:09:06 UTC (07:09:06 -0800 local).	Upper boundary of the evidence: nothing after this time can be seen. Local time is 07:09:06 (UTC-8). The value is the system clock at capture, subject to the inconsistencies listed below.
Supported conclusion	-	-	-	Derived from the rows above (see the Executive Summary and Section 6).	An attempted transfer sequence is supported with Moderate confidence: Chrome holds the URL, F: is mapped to a USB device, and cmd.exe under IEUser recorded a copy command with a success message. Completed transfer and actor remain Low (Section 6).

Timestamp inconsistencies observed in the output

UTC timestamp	Evidence ID	Artefact	Observation	Your treatment
18:02:44 - 18:02:45	E03	Start times of System, smss.exe, csrss.exe, wininit.exe, winlogon.exe, services.exe, lsass.exe, lsm.exe, svchost.exe (552) and VBoxService.exe	Later than the image time (15:09:06) and about three hours after the other services (from 15:02:47).	Not used. These times are later than the image time, which a capture at 15:09:06 cannot contain, and the other services start from 15:02:47. The gap is almost exactly three hours, which suggests a clock or time-zone adjustment during boot (inference); the image cannot confirm the cause. Excluded from the sequence.
15:09:07	E03	dllhost.exe PID 2228 start time	One second after the image time.	One second after the image time. A process can start while the capture is running, so this may reflect the acquisition in progress (inference). It does not affect the findings and is kept out of the sequence.
16:31:39	E05	TimeZoneInformation key last written	Later than the image time.	Later than the image time, so it cannot be a normal write on the same clock; the cause is unexplained and it is not used to date any event. The stored values (Pacific Standard Time, Bias and ActiveTimeBias 480) agree with the local time from imageinfo, so they were used for the UTC conversion.
14:32:53 and 14:49:40	E23	Desktop bag LastWrite; Downloads shell-item Access time	Earlier than the first processes of this boot session (from 15:02:44).	Earlier than the first processes of this boot. Registry data persists between sessions, so these values can come from an earlier session. With the three-hour anomaly they cannot be placed against the 15:0x events, so they are background only and do not date the transfer.

6. Attribution Assessment

Account is not person. The records tie the commands to the IEUser account (token SID ending -1000, E09, E17) and to drive F: in IEUser's profile (E19). They do not identify the human at the keyboard. Anyone who knew or shared the IEUser credentials, or who used the already logged-on console, would leave the same records.

Alternative access route. sshd.exe (PID 2016) was listening on port 22, and the sshd_server profile and sshd account exist (E03, E04, E08, E15, E17). The cmd.exe token shows an interactive console logon, but the records do not rule out that another route produced the activity. This is an unresolved alternative.

Command success is not proven. The console reports "1 file(s) copied." (E07), but filescan returned no file object (E22) and ShellBags contain no F:\ entry (E23). Neither result is proof of absence, and the console text alone does not prove the content on the USB device. The attempt is supported; the completed transfer is not.

Snapshot and timing limits. The image is one capture at 15:09:06 UTC, and cmdscan and consoles carry no per-command times (E02, E06, E07). The commands can only be placed between 15:06:11 and 15:09:06. Events before the processes started, or after the capture, are not visible.

Incomplete or overwritten evidence. The two CLOSED connections to 192.168.56.5:80 have no owning process, and many string hits sit in free memory (E04, E16), so they cannot be tied to Chrome with confidence. The USB instance ID has no serial number, so the device cannot be uniquely identified (E20).

Tool activity during capture. FTK Imager.exe started at 15:07:15, before the capture (E03). The live system was running an acquisition tool while commands and network activity were still possible, which may have altered the volatile state.

Timestamp inconsistencies. The 18:02:44-45 process starts, the 15:09:07 dllhost.exe start, the 16:31:39 TimeZoneInformation write and the earlier ShellBag times do not fit the image time (E03, E05, E23). They were excluded from the sequence, but they reduce confidence in the system clock.

Assessment. The evidence supports account and device presence for IEUser and drive F: and a recorded attempt, with Low confidence in attribution to a person and Low confidence in a completed transfer. Further sources (the disk image, USB device, server logs and logon records) would be needed to move either rating up.

Appendix A: Labelled Screenshots

Each figure shows its evidence ID, plugin, profile and command. Highlighted RECAPTURE notes and SCREENSHOT MISSING boxes mark items to fix before submitting. Redact unrelated personal data.

Figure 1 - Method

Plugin / tool: shell / git | Profile: n/a

mkdir -p ~/DF204_CaseStudy3; cd ~/DF204_CaseStudy3; git --version; git clone https://github.com/volatilityfoundation/volatility.git

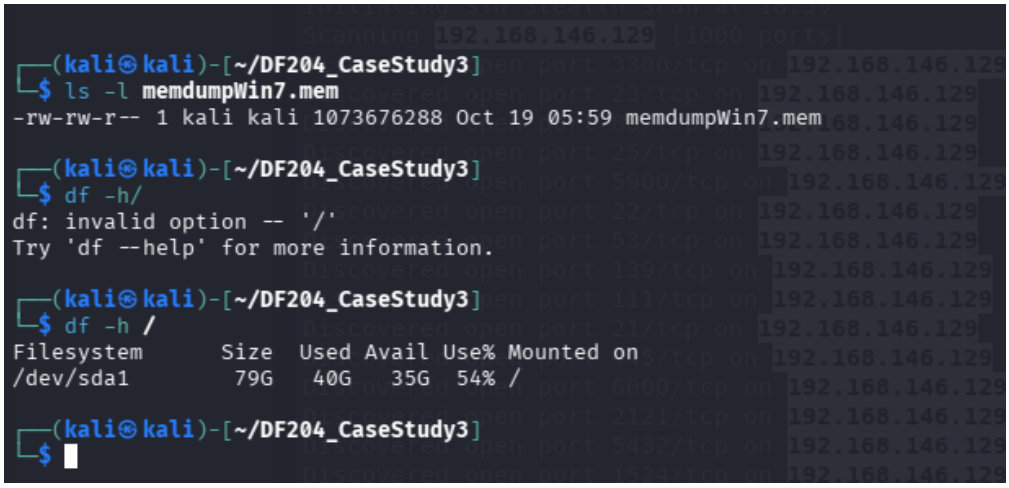


Shows: Case folder created; git 2.51.0; Volatility repository cloned into ~/DF204_CaseStudy3/volatility.

Figure 2 - E01

Plugin / tool: ls / df | Profile: n/a

ls -l memdumpWin7.mem; df -h /



Shows: memdumpWin7.mem present, 1,073,676,288 bytes; 35 GB free on /. (A mistyped "df -h/" returned an error and was re-run correctly.)

Figure 3 - E01

Plugin / tool: sha256sum / chmod / cp | Profile: n/a

sha256sum memdumpWin7.mem | tee original_sha256.txt; chmod 444 memdumpWin7.mem; cp memdumpWin7.mem memdump_working.mem; sha256sum memdump_working.mem | tee working_sha256.txt; ls -l memdumpWin7.mem memdump_working.mem

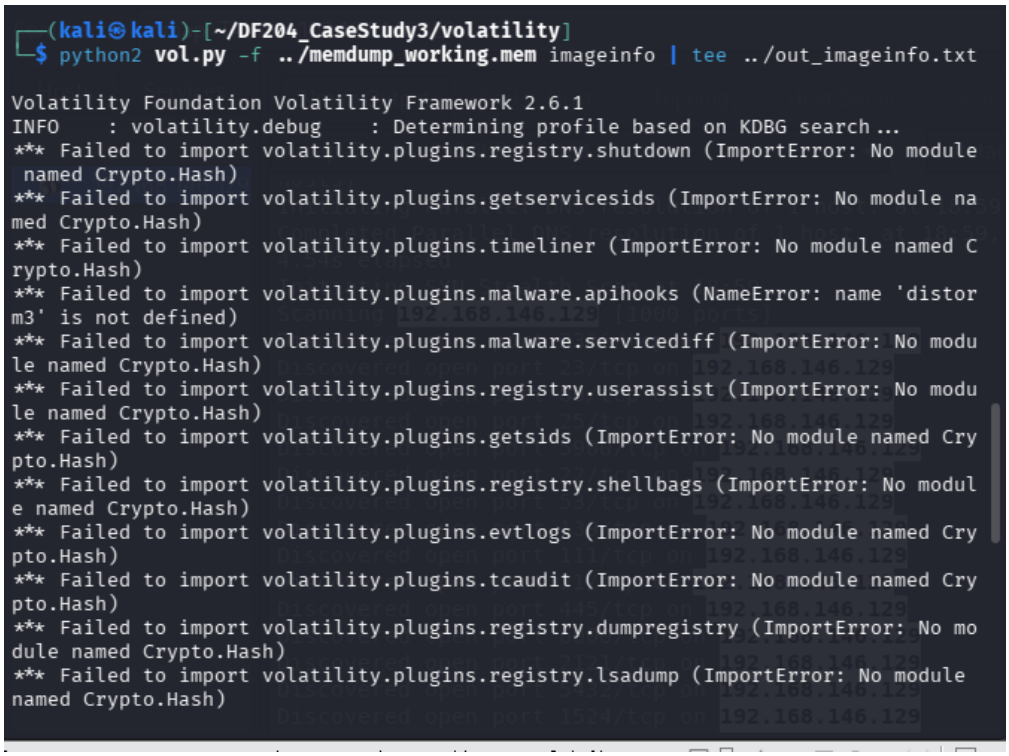


Shows: Identical SHA-256 value for the original and the working copy; both files 1,073,676,288 bytes; original set read-only.

Figure 4 - E02

Plugin / tool: Volatility imageinfo | Profile: n/a (profile search)

python2 vol.py -f ../memdump_working.mem imageinfo | tee ../out_imageinfo.txt

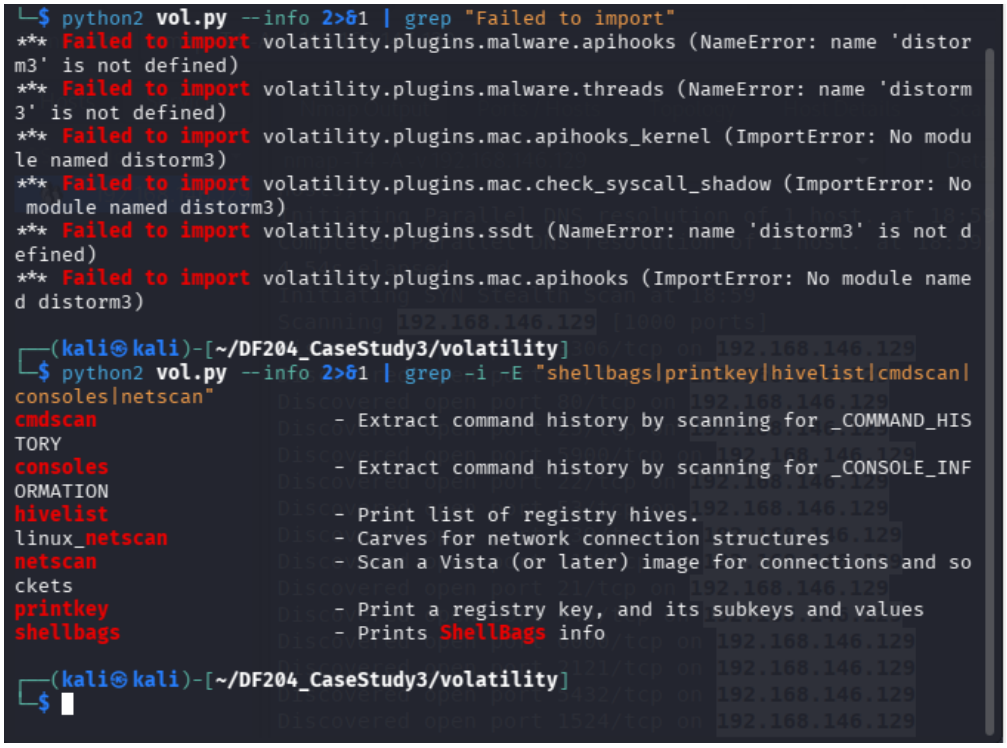


Shows: Volatility Framework 2.6.1 starting its KDBG-based profile search; plugin import warnings (Crypto.Hash, distorm3).

Figure 5 - Method

Plugin / tool: Volatility --info | Profile: n/a

python2 vol.py --info 2>&1 | grep "Failed to import"; python2 vol.py --info 2>&1 | grep -i -E "shellbags|printkey|hivelist|cmdscan|consoles|netscan"

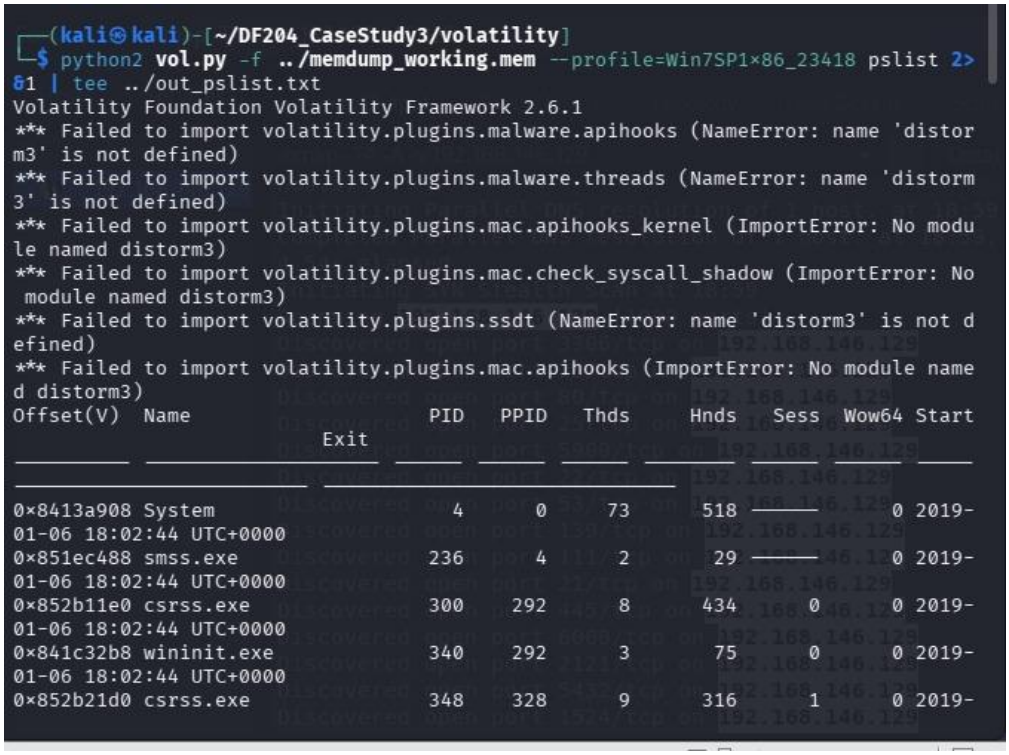


Shows: After installing pycryptodome, only distorm3-related plugins (apihooks, threads, ssdt, mac plugins) fail to import; cmdscan, consoles, hivelist, netscan, printkey and shellbags are available.

Figure 6 - E03

Plugin / tool: Volatility pslist | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 pslist 2>&1 | tee ../out_pslist.txt



```
┌──(kali㉿kali)-[~/DF204_CaseStudy3/volatility]
└─$ python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 pslist 2>
&1 | tee ../out_pslist.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distor
m3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm
3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No modu
le named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No
 module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not d
efined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module name
d distorm3)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start
                              Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ -----
---------- ------------------------------

0x8413a908 System                  4      0     73      518 ------      0 2019-
01-06 18:02:44 UTC+0000
0x851ec488 smss.exe              236      4      2       29 ------      0 2019-
01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe             300    292      8      434      0      0 2019-
01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe           340    292      3       75      0      0 2019-
01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe             348    328      9      316      1      0 2019-
```

Shows: First rows of the process list: System (PID 4), smss.exe (236), csrss.exe (300, 348), wininit.exe (340), all with start time 2019-01-06 18:02:44 UTC.

Figure 7 - E03

Plugin / tool: Volatility psscan | Profile: Win7SP1x86_23418

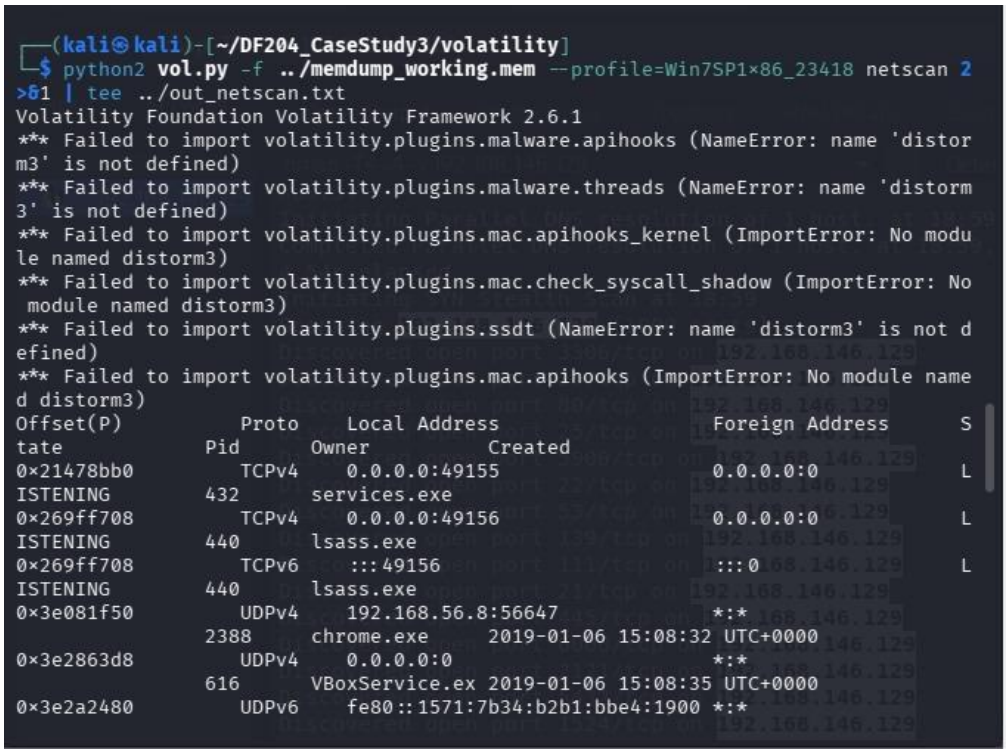
python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 psscan 2>/dev/null | grep -E "Offset|chrome|cmd.exe|conhost|FTK|explorer" | tee ../out_psscan.txt



Figure 8 - E04

Plugin / tool: Volatility netscan | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 netscan 2>&1 | tee ../out_netscan.txt

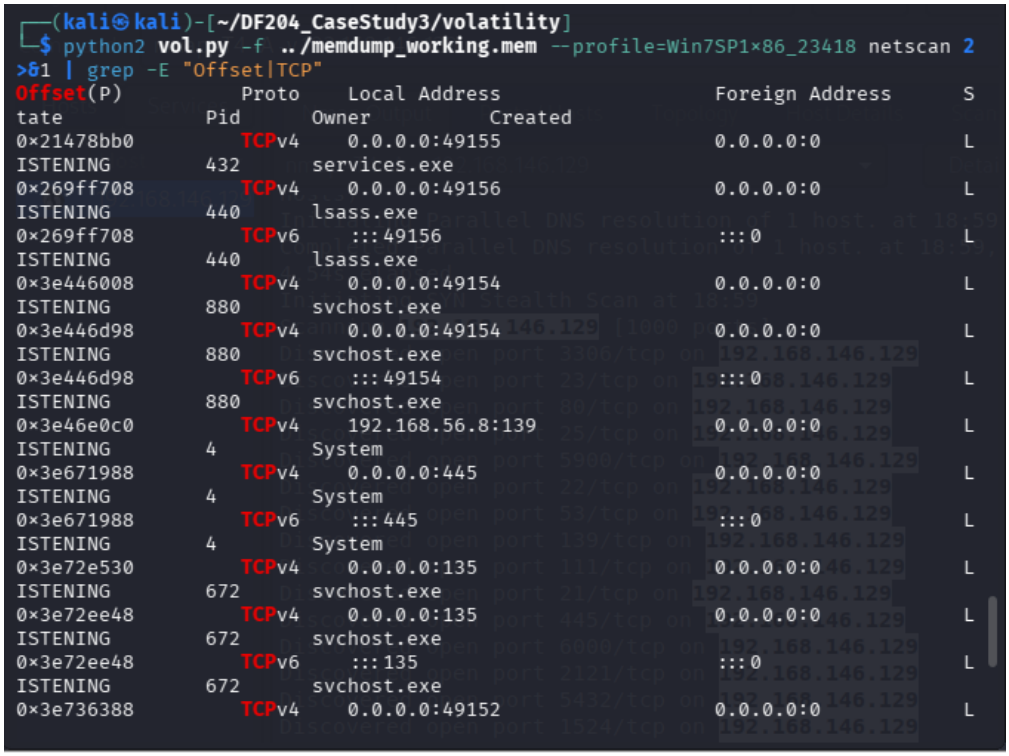


Shows: First socket records: listeners, and a chrome.exe (PID 2388) UDPv4 socket 192.168.56.8:56647 created 2019-01-06 15:08:32 UTC.

Figure 9 - E04

Plugin / tool: Volatility netscan | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 netscan 2>&1 | grep -E "Offset|TCP"



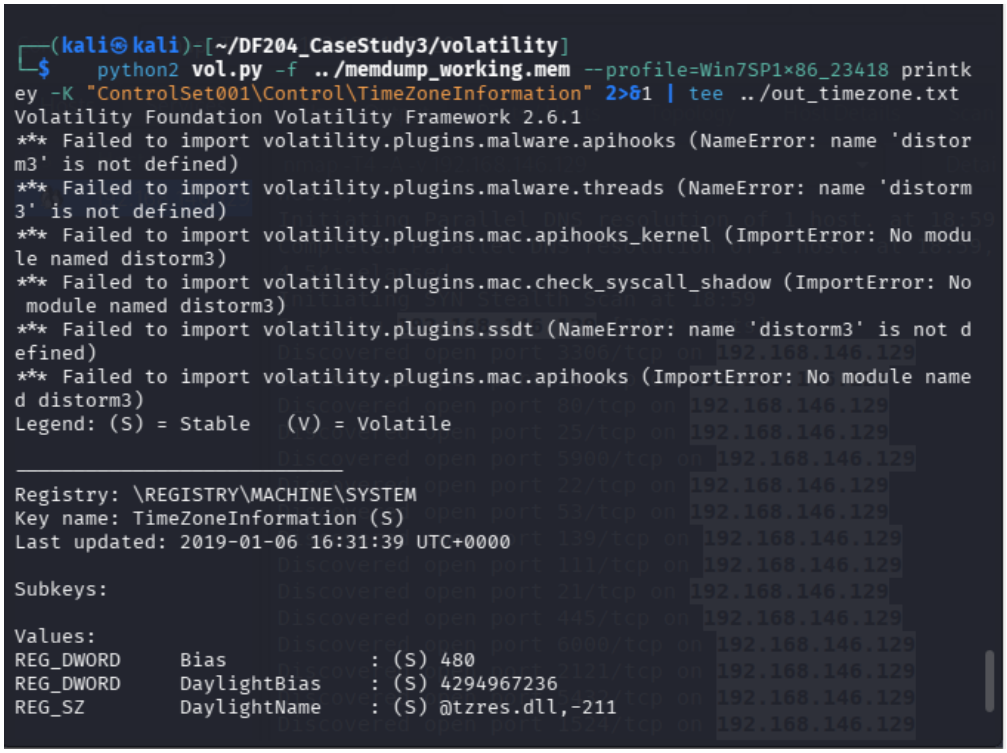
```
┌──(kali㉿kali)-[~/DF204_CaseStudy3/volatility]
└─$ python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 netscan 2
>&1 | grep -E "Offset|TCP"
Offset(P)          Proto    Local Address                  Foreign Address      S
tate            Pid      Owner          Created
0x21478bb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0            L
ISTENING        432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0            L
ISTENING        440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                 L
ISTENING        440      lsass.exe
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0            L
ISTENING        880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0            L
ISTENING        880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                 L
ISTENING        880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0            L
ISTENING        4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0            L
ISTENING        4        System
0x3e671988         TCPv6    :::445                         :::0                 L
ISTENING        4        System
0x3e72e530         TCPv4    0.0.0.0:135                    0.0.0.0:0            L
ISTENING        672      svchost.exe
0x3e72ee48         TCPv4    0.0.0.0:135                    0.0.0.0:0            L
ISTENING        672      svchost.exe
0x3e72ee48         TCPv6    :::135                         :::0                 L
ISTENING        672      svchost.exe
0x3e736388         TCPv4    0.0.0.0:49152                  0.0.0.0:0            L
```

Shows: Header and the first TCP listener records.

Figure 10 - E05

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\TimeZoneInformation" 2>&1 | tee ../out_timezone.txt



Shows: SYSTEM hive TimeZoneInformation key last updated 2019-01-06 16:31:39 UTC; Bias 480; DaylightBias; DaylightName.

Figure 11 - E06

Plugin / tool: Volatility cmdscan | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 cmdscan 2>&1 | tee ../out_cmdscan.txt

```
└─$ python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 cmdscan 2
>&1 | tee ../out_cmdscan.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distor
m3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm
3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No modu
le named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No
 module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not d
efined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module name
d distorm3)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
```

Figure 12 - E07

Plugin / tool: Volatility consoles | Profile: Win7SP1x86_23418

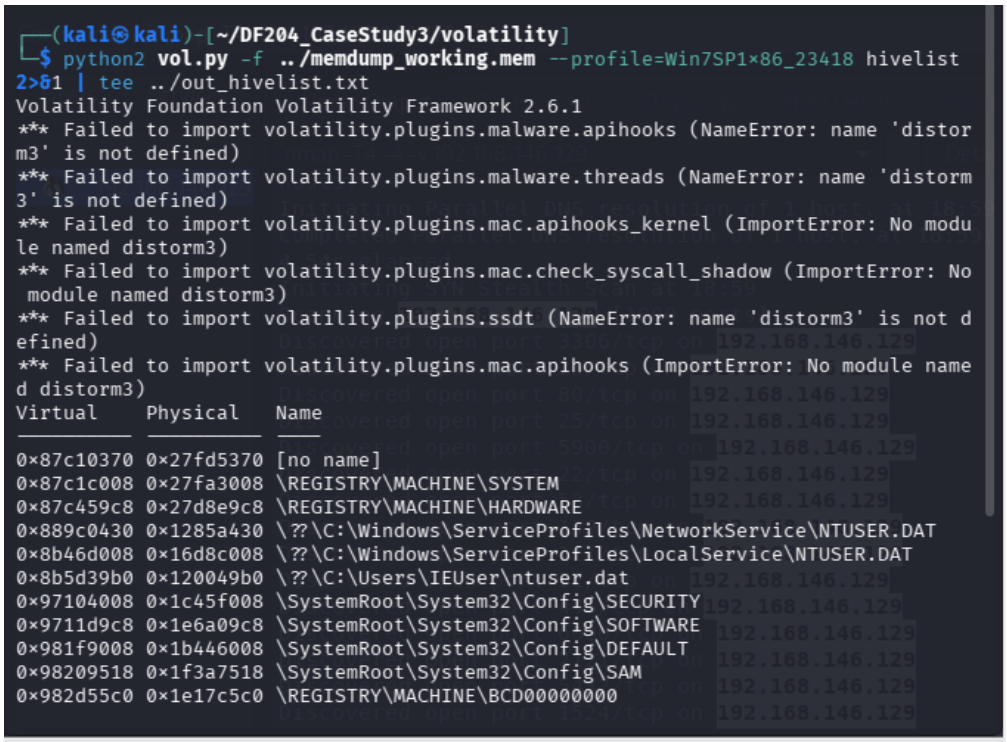
python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 consoles 2>&1 | tee ../out_consoles.txt

```
┌──(kali㉿kali)-[~/DF204_CaseStudy3/volatility]
└─$ python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 consoles
2>&1 | tee ../out_consoles.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distor
m3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm
3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No modu
le named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No
 module named distorm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not d
efined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module name
d distorm3)
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
```

Figure 13 - E08

Plugin / tool: Volatility hivelist | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 hivelist 2>&1 | tee ../out_hivelist.txt

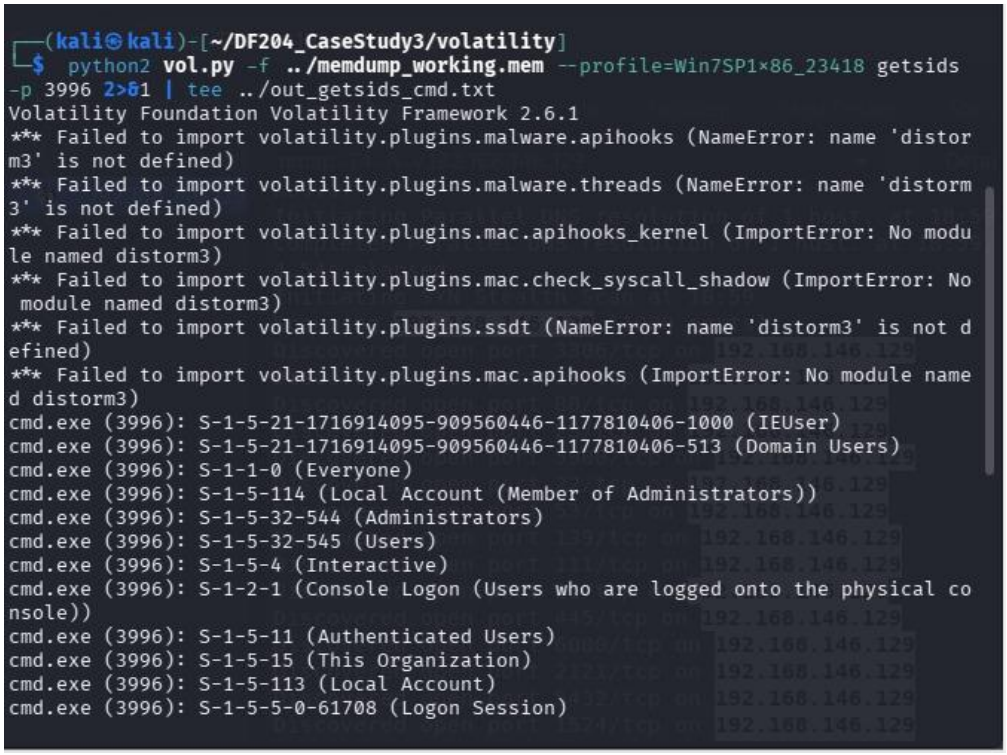


Shows: Registry hive table: SYSTEM, HARDWARE, service-profile NTUSER.DAT hives, \??\C:\Users\IEUser\ntuser.dat, SECURITY, SOFTWARE, DEFAULT, SAM and BCD (visible portion).

Figure 14 - E09

Plugin / tool: Volatility getsids | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 getsids -p 3996 2>&1 | tee ../out_getsids_cmd.txt

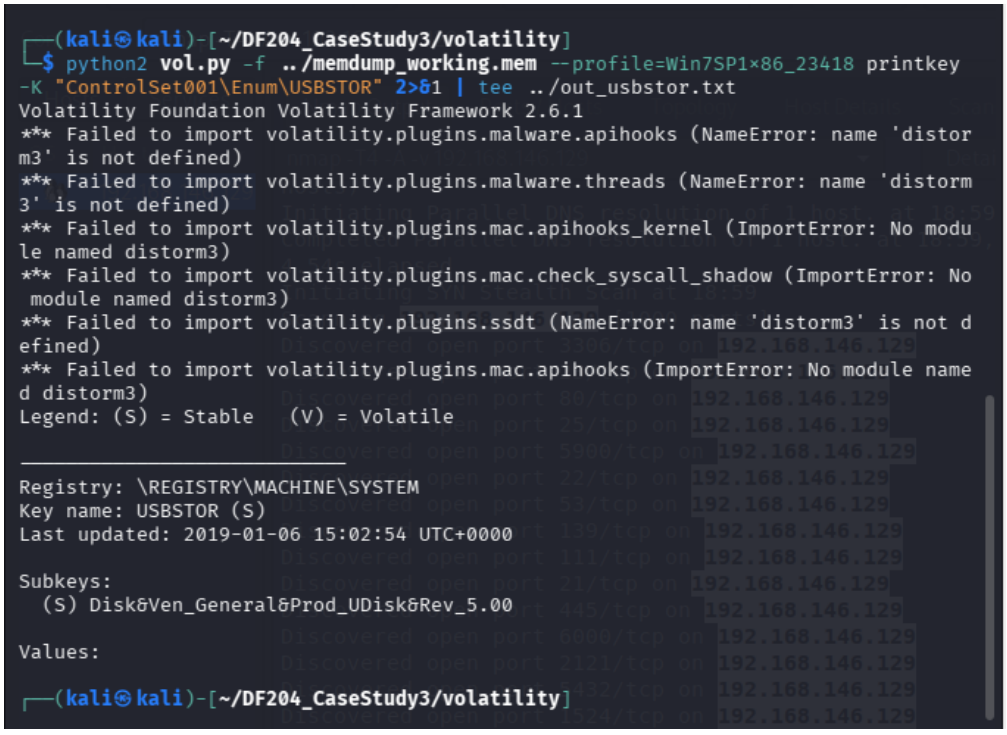


Shows: Token SIDs of cmd.exe (PID 3996): IEUser (S-1-5-21-...-1000), Domain Users (-513), Administrators, Interactive and Console Logon groups.

Figure 15 - E10

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\USBSTOR" 2>&1 | tee ../out_usbstor.txt

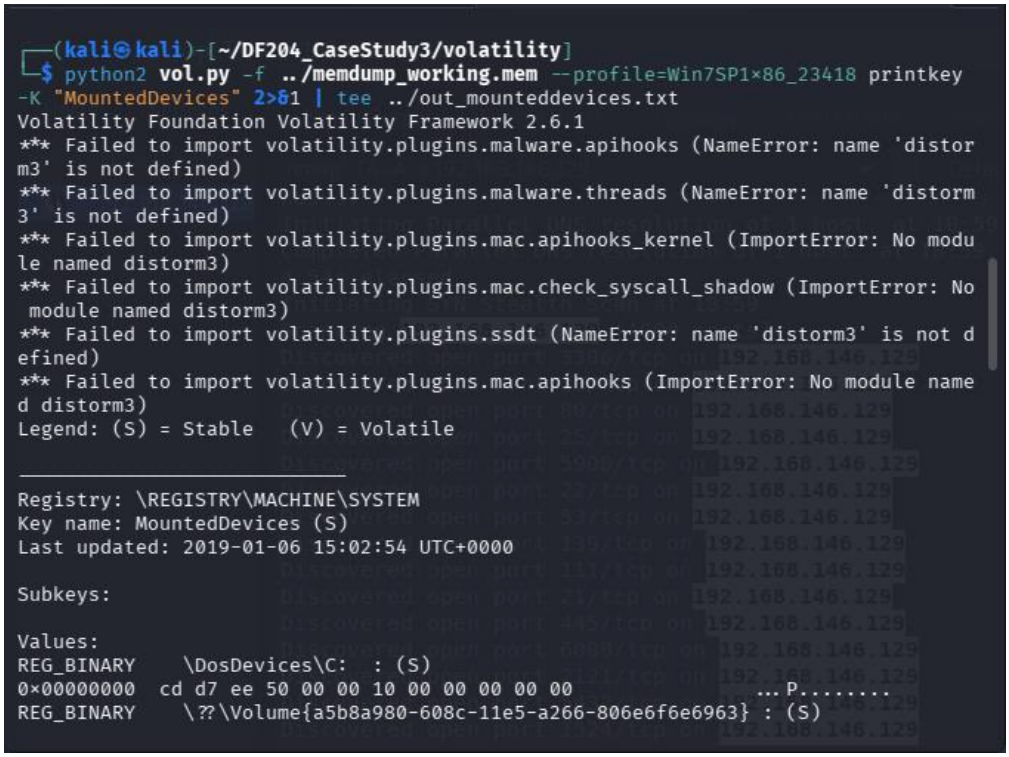


Shows: SYSTEM hive; USBSTOR key last updated 2019-01-06 15:02:54 UTC; one subkey, Disk&Ven_General&Prod_UDisk&Rev_5.00.

Figure 16 - E11

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "MountedDevices" 2>&1 | tee ../out_mounteddevices.txt

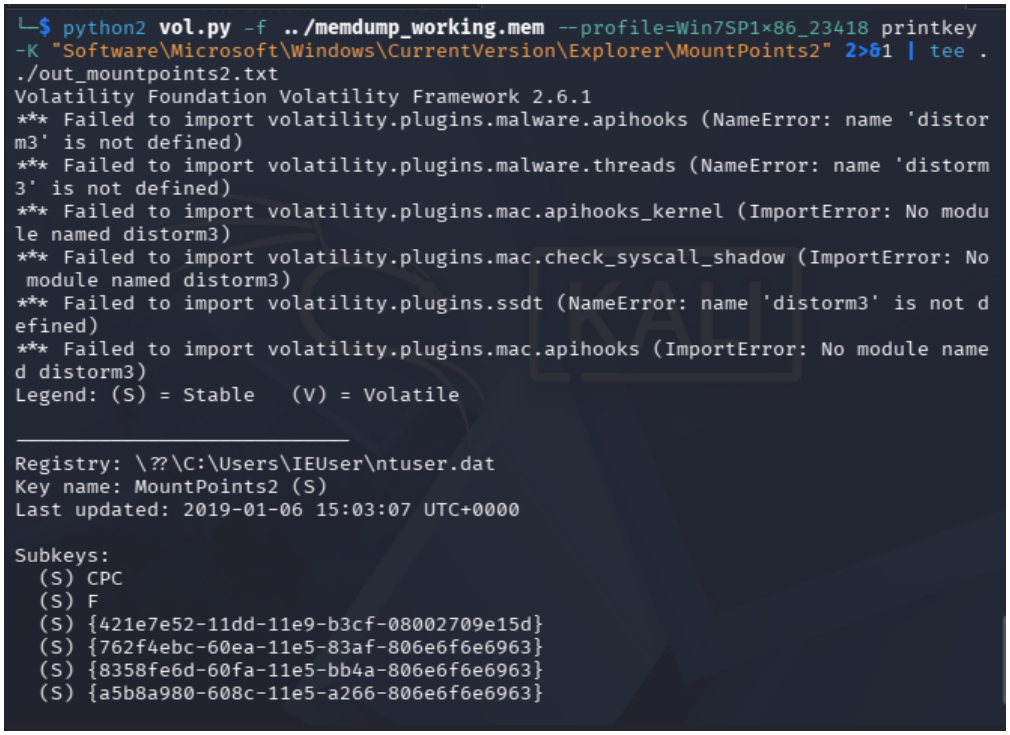


Shows: MountedDevices key last updated 2019-01-06 15:02:54 UTC; start of the values list (\DosDevices\C:).

Figure 17 - E12

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2" 2>&1 | tee ../out_mountpoints2.txt



Shows: IEUser ntuser.dat MountPoints2 key last updated 2019-01-06 15:03:07 UTC; subkeys include CPC, F and volume GUIDs including {421e7e52-11dd-11e9-b3cf-08002709e15d}.

Figure 18 - Limit

Plugin / tool: Volatility yarascan | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 yarascan -Y "192.168.56.5" 2>&1 | tee ../out_yara_ip.txt

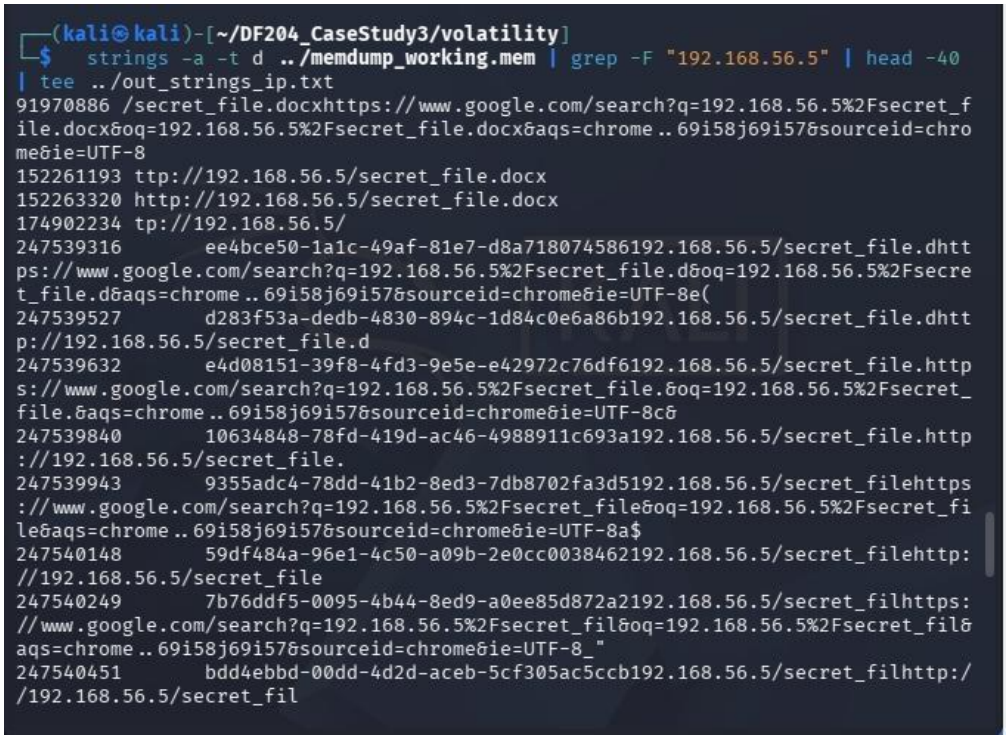


Shows: Plugin could not run: Yara is not installed in this environment. The address search was done with strings instead (E13, E16).

Figure 19 - E13

Plugin / tool: strings + grep (file offsets) | Profile: n/a

strings -a -t d ../memdump_working.mem | grep -F "192.168.56.5" | head -40 | tee ../out_strings_ip.txt

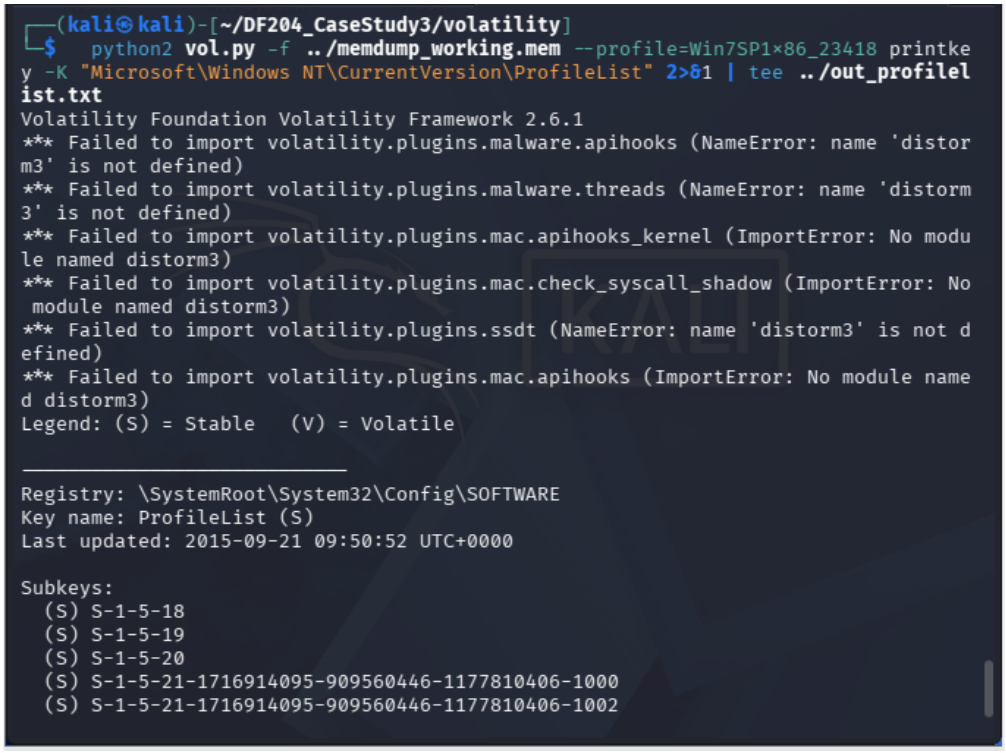


Shows: File offsets of strings referencing 192.168.56.5/secret_file.docx, with Google search URLs for the same text.

Figure 20 E14

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList" 2>&1 | tee ../out_profilelist.txt

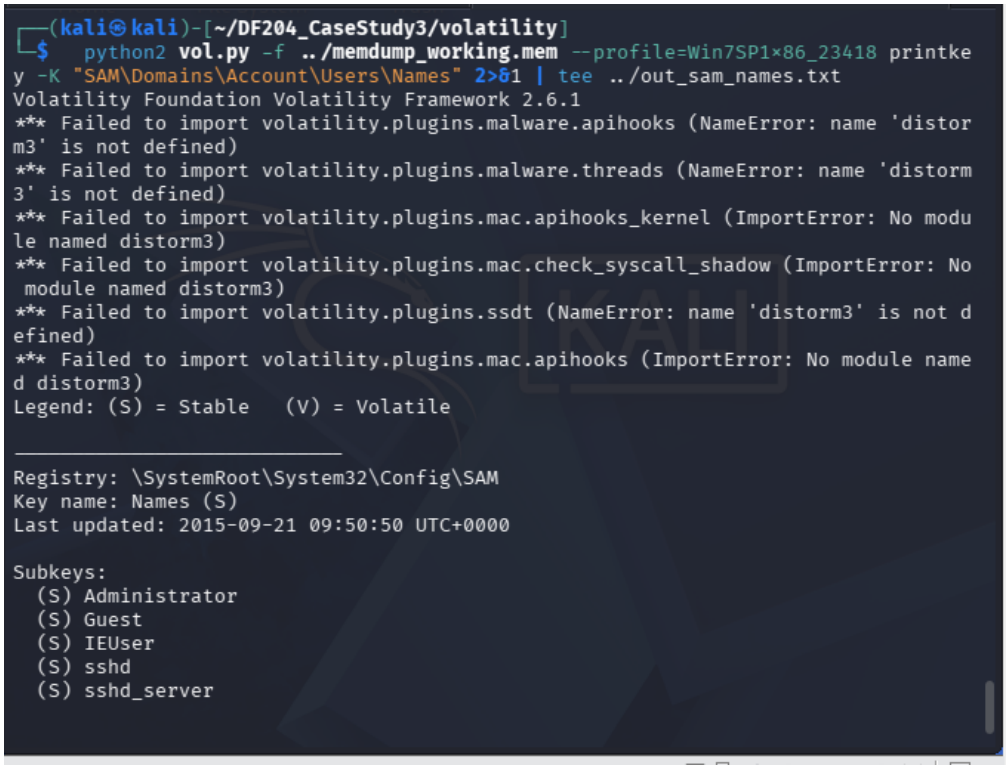


Shows: SOFTWARE hive ProfileList key (last updated 2015-09-21 09:50:52 UTC) with subkeys S-1-5-18, -19, -20, ...-1000 and ...-1002.

Figure 21 - E15

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "SAM\Domains\Account\Users\Names" 2>&1 | tee ../out_sam_names.txt

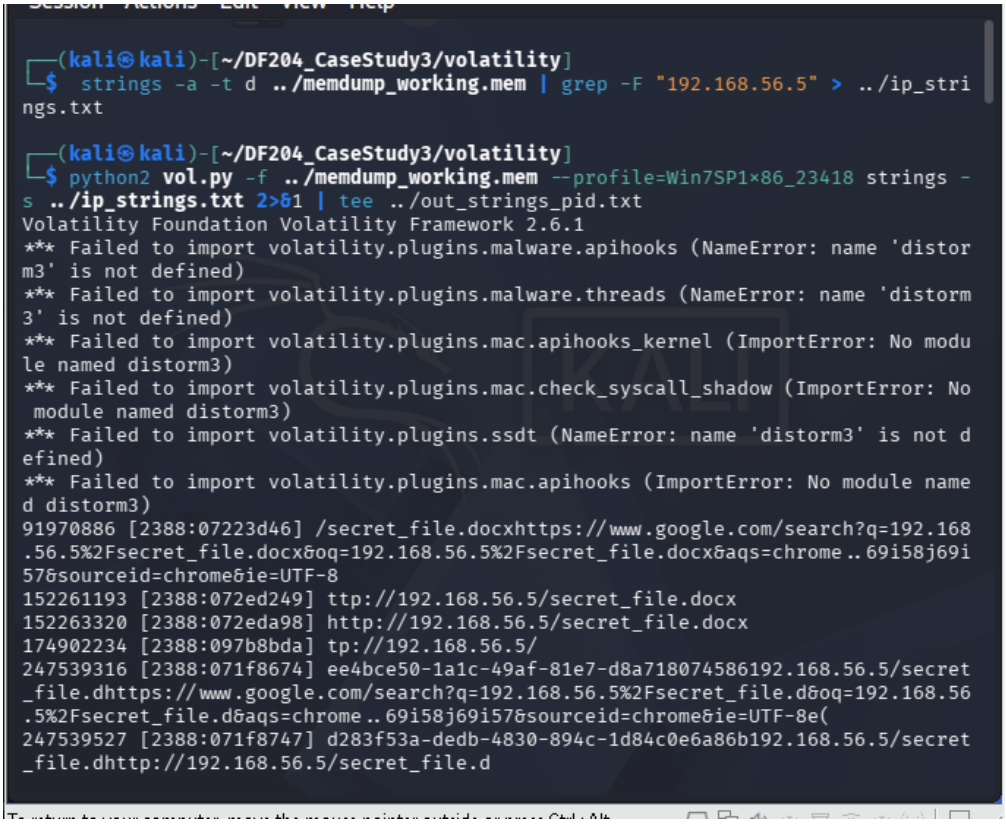


Shows: SAM Names key (last updated 2015-09-21 09:50:50 UTC): Administrator, Guest, IEUser, sshd, sshd_server.

Figure 22 - E16

Plugin / tool: Volatility strings | Profile: Win7SP1x86_23418

strings -a -t d ../memdump_working.mem | grep -F "192.168.56.5" > ../ip_strings.txt; python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 strings -s ../ip_strings.txt 2>&1 | tee ../out_strings_pid.txt

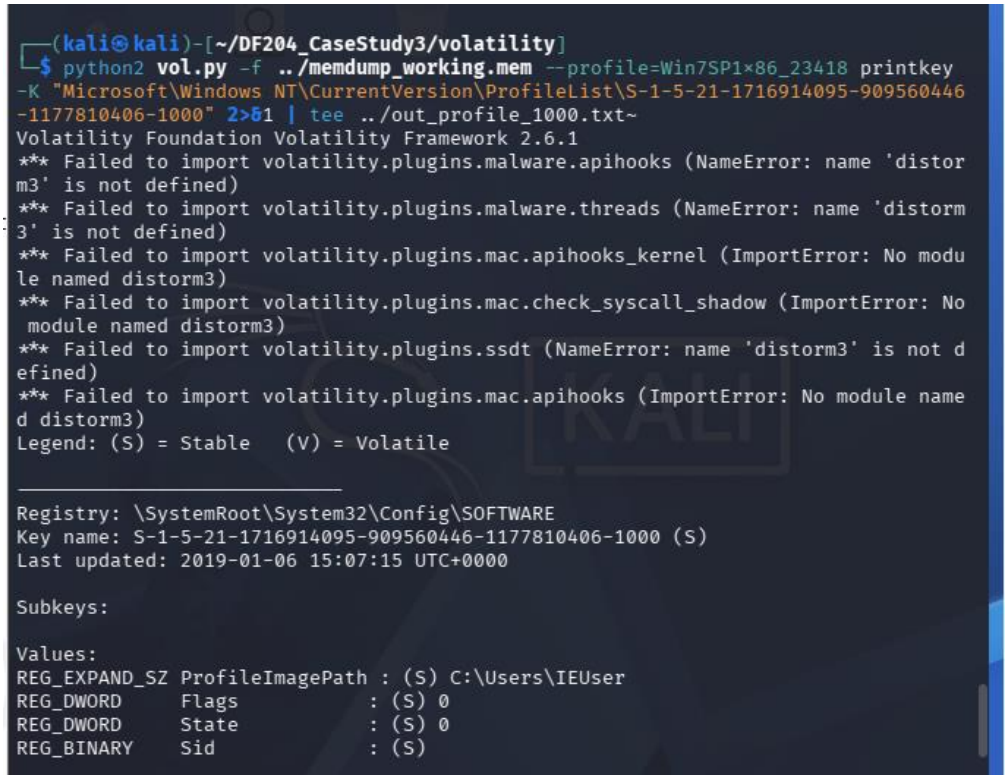


Shows: Strings mapped to process and virtual address, e.g. 152261193 [2388:072ed249] for the URL http://192.168.56.5/secret_file.docx.

Figure 23 - E17

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1000" 2>&1 | tee ../out_profile_1000.txt~

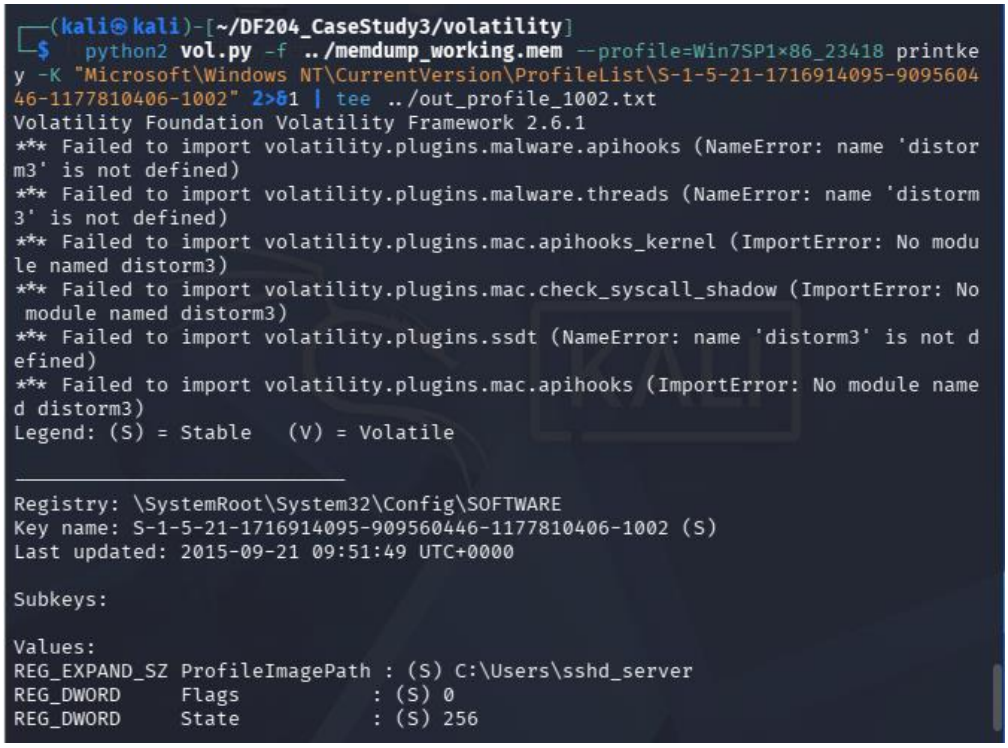


Shows: SID ...-1000: ProfileImagePath C:\Users\IEUser; key last updated 2019-01-06 15:07:15 UTC.

Figure 24 - E17

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002" 2>&1 | tee ../out_profile_1002.txt

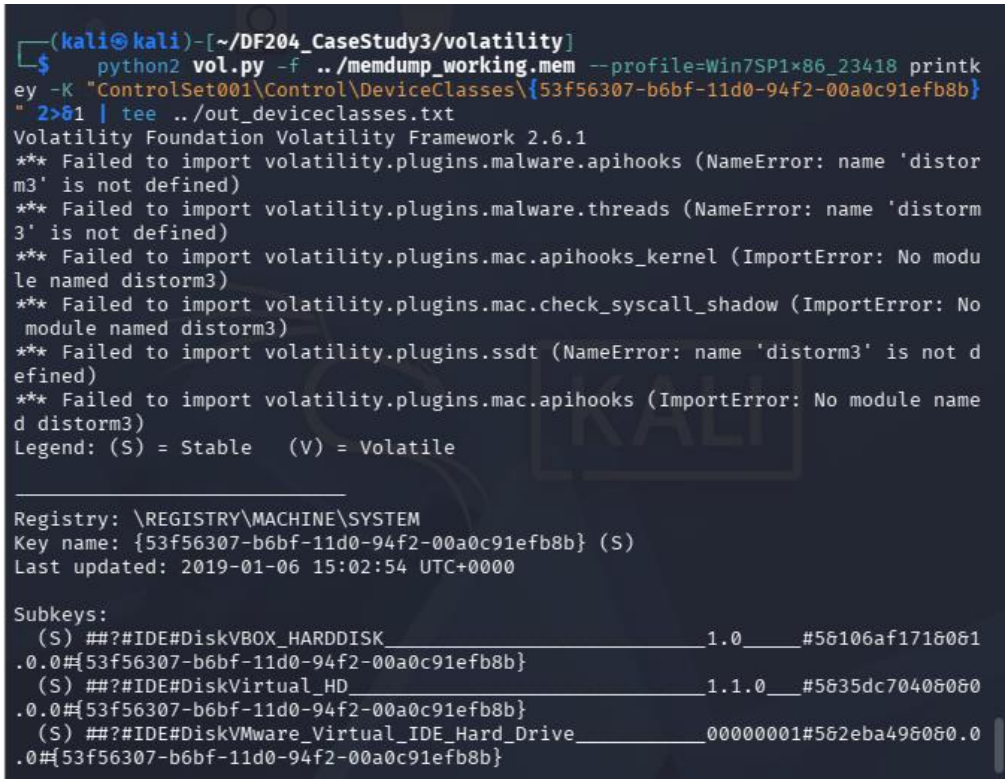


Shows: SID ...-1002: ProfileImagePath C:\Users\sshd_server; key last updated 2015-09-21 09:51:49 UTC.

Figure 25 - E18

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}" 2>&1 | tee ../out_deviceclasses.txt

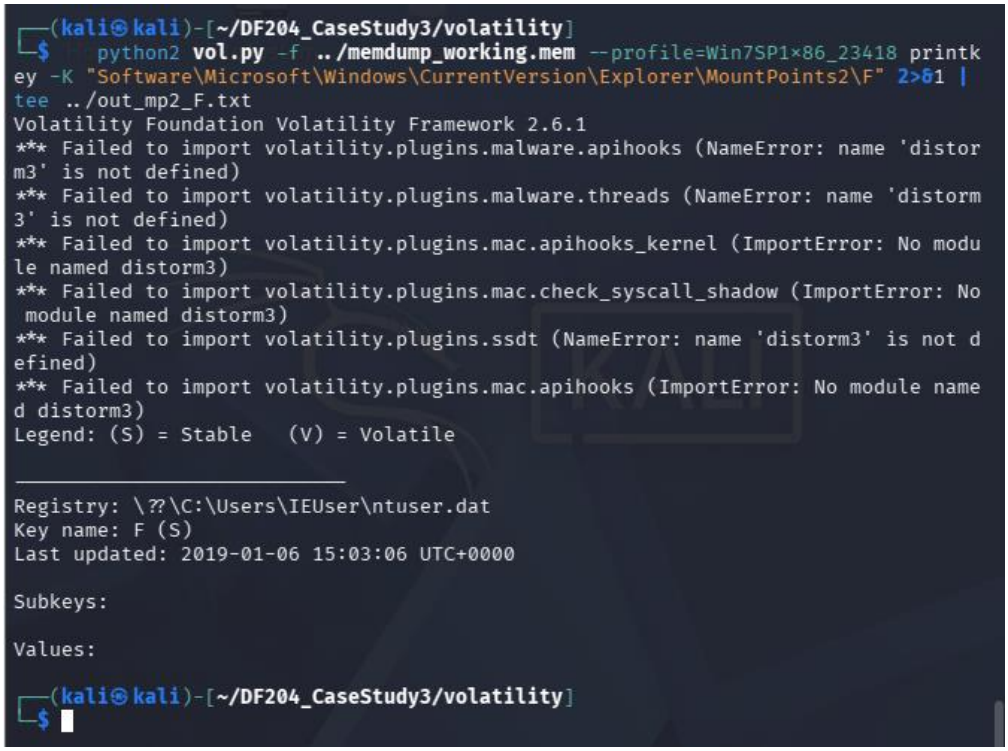


Shows: DeviceClasses disk-interface key last updated 2019-01-06 15:02:54 UTC; the first listed interfaces are IDE disks.

Figure 26 - E19

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F" 2>&1 | tee ../out_mp2_F.txt

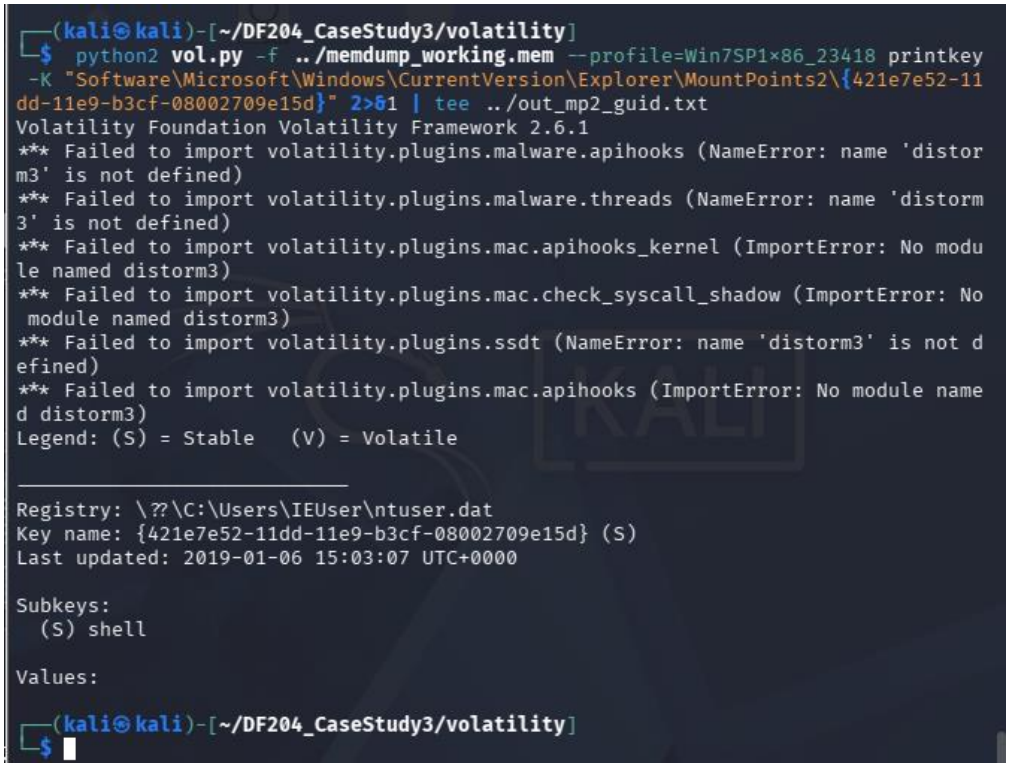


Shows: IEUser ntuser.dat MountPoints2\F last updated 2019-01-06 15:03:06 UTC.

Figure 27 - E19

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\{421e7e52-11dd-11e9-b3cf-08002709e15d}" 2>&1 | tee ../out_mp2_guid.txt

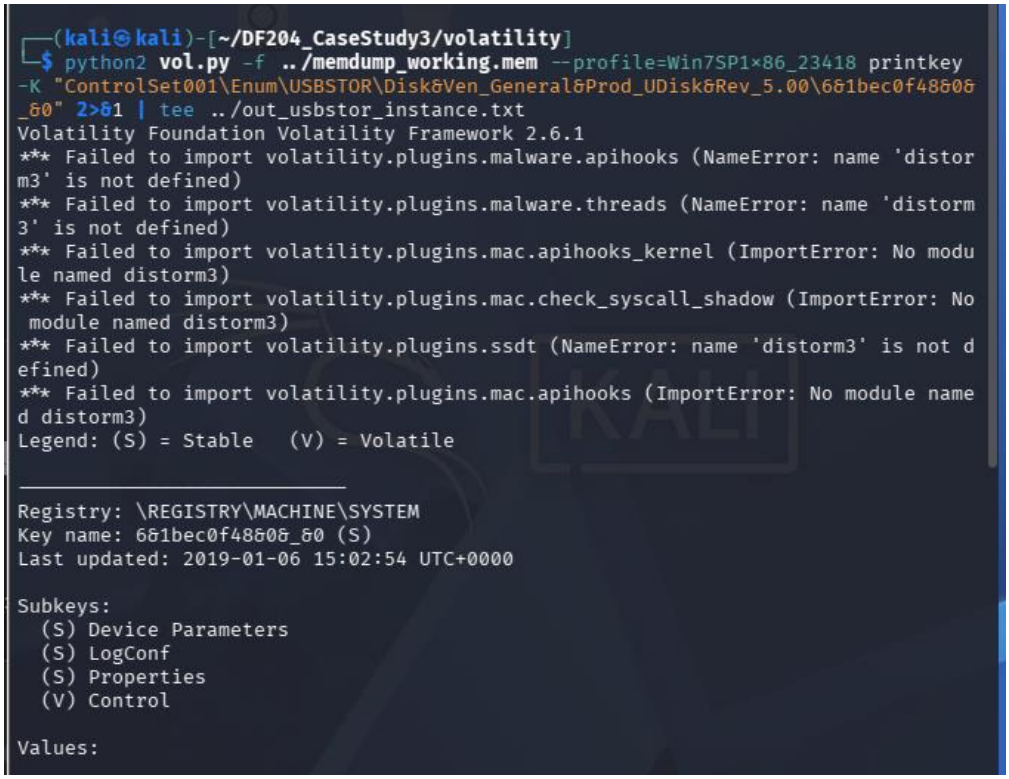


Shows: MountPoints2\{421e7e52-11dd-11e9-b3cf-08002709e15d} last updated 2019-01-06 15:03:07 UTC, with a "shell" subkey.

Figure 28 - E20

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0" 2>&1 | tee ../out_usbstor_instance.txt

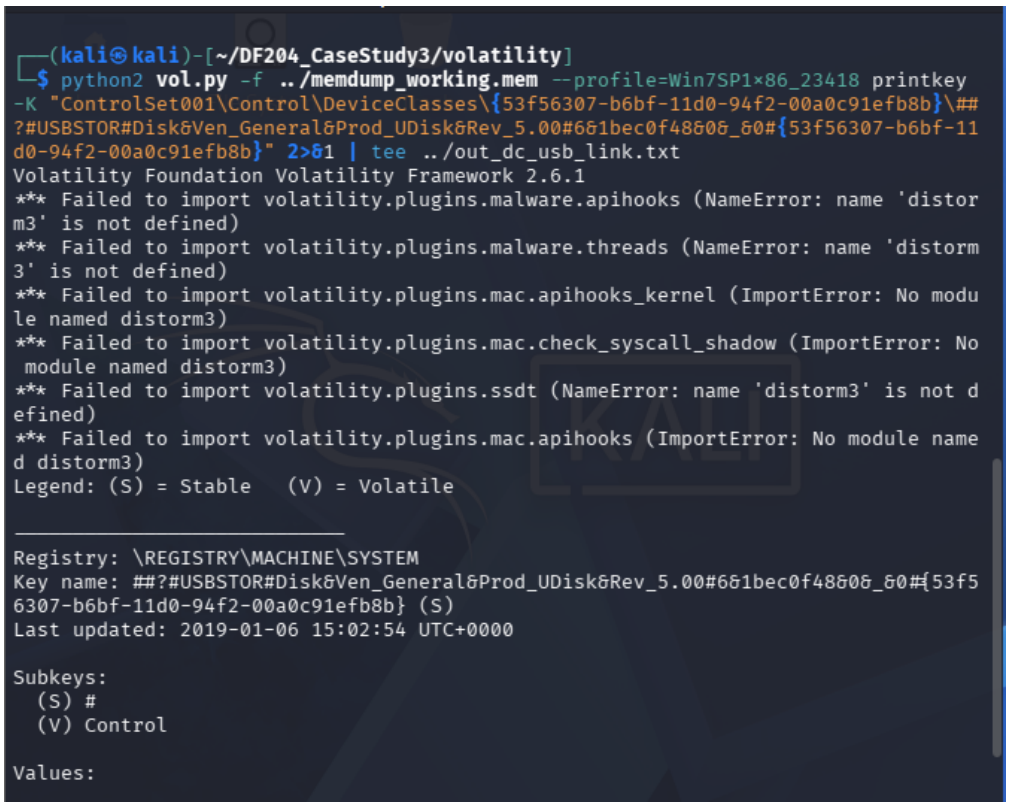


Shows: USBSTOR instance key 6&1bec0f48&0&_&0 last updated 2019-01-06 15:02:54 UTC; subkeys Device Parameters, LogConf, Properties, Control.

Figure 29 - E21

Plugin / tool: Volatility printkey | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}" 2>&1 | tee ../out_dc_usb_link.txt

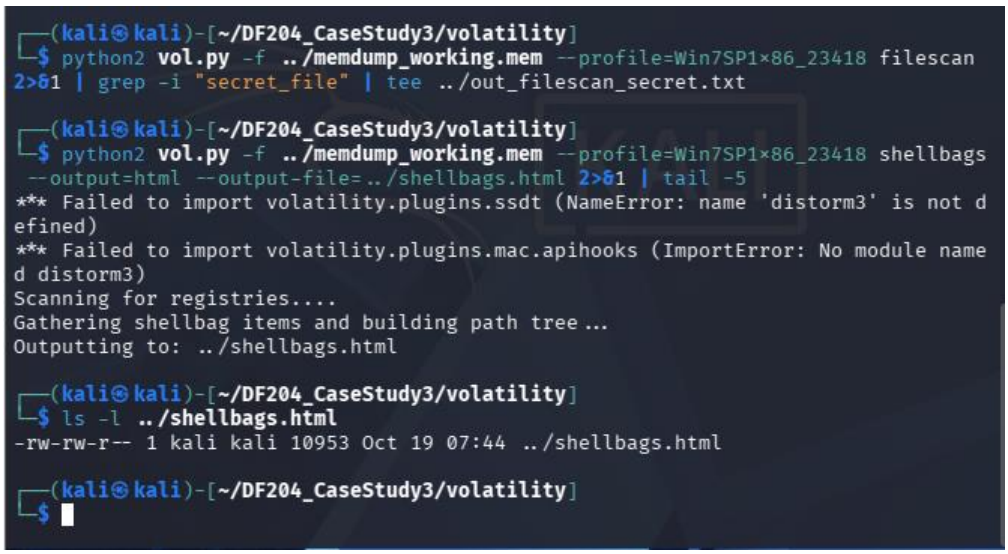


Shows: DeviceClasses interface link for the USBSTOR device, last updated 2019-01-06 15:02:54 UTC, with subkeys "#" and Control.

Figure 30 - E22, E23

Plugin / tool: Volatility filescan; shellbags | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 filescan 2>&1 | grep -i "secret_file" | tee ../out_filescan_secret.txt; python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 shellbags --output=html --output-file=../shellbags.html 2>&1 | tail -5; ls -l ../shellbags.html



Shows: The filescan search for "secret_file" returned no output; ShellBags HTML report generated (10,953 bytes).

Figure 31 - E22, E23

Plugin / tool: Volatility filescan; shellbags | Profile: Win7SP1x86_23418

python2 vol.py -f ../memdump_working.mem --profile=Win7SP1x86_23418 filescan 2>&1 | grep -i -E "secret|Downloads.*docx" | tee ../out_filescan_wide.txt; firefox ../shellbags.html &; sed 's/<[^>]*>/ /g' ../shellbags.html | tr -s ' ' | grep -v '^ *$' | tee ../shellbags_text.txt



Shows: The wider filescan search also returned no output; shellbags.html opened and its data converted to text (first Desktop-bag rows shown).

Figure 32 - E23

Plugin / tool: Volatility shellbags (HTML view) | Profile: Win7SP1x86_23418

firefox ../shellbags.html

Appendix B: Evidence Log

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01	sha256sum, chmod, cp	~/DF204_CaseStudy3/memdumpWin7.mem; memdump_working.mem	SHA-256 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e for both original and working copy; 1,073,676,288 bytes; original set read-only (chmod 444).	Fig. 2, Fig. 3

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E02	Volatility imageinfo	KDBG search; KDBG 0x82972c68, DTB 0x185000	Suggested profiles Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86; selected Win7SP1x86_23418; image date/time 2019-01-06 15:09:06 UTC (07:09:06 -0800 local).	Fig. 4 App. C1
E03	Volatility pslist (Win7SP1x86_23418)	Process list	explorer.exe PID 2524 start 15:02:54; chrome.exe PID 2388 (PPID 2524) start 15:04:30; cmd.exe PID 3996 (PPID 2524) and conhost.exe PID 3988 start 15:06:11; FTK Imager.exe PID 2596 start 15:07:15; sshd.exe PID 2016; earliest processes (System, smss.exe ...) show start 18:02:44-45.	Fig. 6, Fig. 7 App. C2
E04	Volatility netscan (Win7SP1x86_23418)	Offsets 0x3e043b10, 0x3fcba008, 0x3e081f50	TCPv4 192.168.56.8:49177 and :49178 to 192.168.56.5:80, state CLOSED, PID -1, no owner; chrome.exe PID 2388 UDPv4 192.168.56.8:56647 created 15:08:32; sshd.exe PID 2016 listening on port 22.	Fig. 8, Fig. 9 App. C3
E05	Volatility printkey (Win7SP1x86_23418)	SYSTEM: ControlSet001\Control\TimeZoneInformation	Pacific Standard Time; Bias and ActiveTimeBias 480 (UTC-8); key last written 2019-01-06 16:31:39 UTC.	Fig. 10 App. C4
E06	Volatility cmdscan (Win7SP1x86_23418)	conhost.exe PID 3988; CommandHistory 0x110448 (cmd.exe)	CommandCount 3: Cmd #0 ipconfig (0x10dc40); #1 cd Downloads (0x107f90); #2 copy secret_file.docx F: (0x1147a0). conhost.exe PID 2000 (sshd.exe): CommandCount 0.	Fig. 11 App. C5
E07	Volatility consoles (Win7SP1x86_23418)	conhost.exe PID 3988 / cmd.exe PID 3996 screen buffer	Prompt C:\Users\IEUser; ipconfig output (IPv4 192.168.56.8); C:\Users\IEUser\Downloads>copy secret_file.docx F: followed by "1 file(s) copied."	Fig. 12 App. C6
E08	Volatility hivelist (Win7SP1x86_23418)	Hive offsets	IEUser ntuser.dat 0x8b5d39b0; IEUser UsrClass.dat 0xa620b008; sshd_server ntuser.dat 0xa65269c8; SYSTEM 0x87c1c008; SOFTWARE 0x9711d9c8; SAM 0x98209518.	Fig. 13 App. C8
E09	Volatility getsids -p 3996 (Win7SP1x86_23418)	cmd.exe token	S-1-5-21-1716914095-909560446-1177810406-1000 (IEUser); member of Administrators, Interactive and Console Logon groups.	Fig. 14 App. C7
E10	Volatility printkey (Win7SP1x86_23418)	SYSTEM: ControlSet001\Enum\USBSTOR	Subkey Disk&Ven_General&Prod_UDisk&Rev_5.00; key last written 2019-01-06 15:02:54 UTC.	Fig. 15 App. C9

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E11	Volatility printkey (Win7SP1x86_23418)	SYSTEM: MountedDevices	\DosDevices\F: maps to _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-...} (decoded from hex); volume {421e7e52-11dd-11e9-b3cf-08002709e15d} maps to the same string; \DosDevices\E: is a VirtualBox CD-ROM; key last written 15:02:54 UTC.	Fig. 16 App. C9
E12	Volatility printkey (Win7SP1x86_23418)	IEUser ntuser.dat: ...\Explorer\MountPoints2	Subkeys CPC, F, {421e7e52-11dd-11e9-b3cf-08002709e15d}, {762f4ebc-...}, {8358fe6d-...}, {a5b8a980-...}, {a5b8a983-...}; key last written 15:03:07 UTC.	Fig. 17 App. C9
E13	strings -a -t d + grep -F (file offsets)	Offsets e.g. 152261193, 152263320, 568756597	URL http://192.168.56.5/secret_file.docx; progressively shorter fragments beside Google search URLs; cache header at 568756597 with Last-Modified Fri, 04 Jan 2019 16:05:12 GMT and a wordprocessingml.document content type (same record also contains the string 192.128.56.5).	Fig. 19 App. C10
E14	Volatility printkey (Win7SP1x86_23418)	SOFTWARE: Microsoft\Windows NT\CurrentVersion\ProfileList	Subkeys S-1-5-18, -19, -20, S-1-5-21-1716914095-909560446-1177810406-1000 and -1002; key last written 2015-09-21 09:50:52 UTC.	Fig. 20 App. C11
E15	Volatility printkey (Win7SP1x86_23418)	SAM: SAM\Domains\Account\Users\Names	Administrator, Guest, IEUser, sshd, sshd_server; key last written 2015-09-21 09:50:50 UTC.	Fig. 21 App. C11
E16	Volatility strings -s (Win7SP1x86_23418)	ip_strings.txt mapped to PID and virtual address	Many hits mapped to chrome.exe PID 2388 (e.g. 152263320 [2388:072eda98]; 568756597 [2388:0720c975]; 799213234 [2388:060bc6b2], an omnibox suggestion request); many others [FREE MEMORY:-1]; two hits mapped to [1228:... 2388:...].	Fig. 22 App. C10
E17	Volatility printkey (Win7SP1x86_23418)	SOFTWARE: ...\ProfileList\<SID>	-1000: ProfileImagePath C:\Users\IEUser, last written 2019-01-06 15:07:15 UTC; -1002: C:\Users\sshd_server, last written 2015-09-21 09:51:49 UTC.	Fig. 23, Fig. 24 App. C11
E18	Volatility printkey (Win7SP1x86_23418)	SYSTEM: ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}	Key last written 15:02:54 UTC; lists disk interfaces including ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{...}.	Fig. 25 App. C9

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E19	Volatility printkey (Win7SP1x86_23418)	IEUser ntuser.dat: ...\MountPoints2\F and \{421e7e52-11dd-11e9-b3cf-08002709e15d}	MountPoints2\F last written 2019-01-06 15:03:06 UTC; volume GUID key (subkey "shell") last written 15:03:07 UTC.	Fig. 26, Fig. 27 App. C9
E20	Volatility printkey (Win7SP1x86_23418)	SYSTEM: ...\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0	FriendlyName General UDisk USB Device; ContainerID {421e7e37-11dd-11e9-b3cf-806e6f6e6963}; HardwareID USBSTOR\DiskGeneral_UDisk_____5.00 (and variants); key last written 15:02:54 UTC; second character of the instance ID is "&".	Fig. 28 App. C9
E21	Volatility printkey (Win7SP1x86_23418)	SYSTEM: DeviceClasses\{53f56307-...}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{...}	DeviceInstance USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0; key last written 15:02:54 UTC.	Fig. 29 App. C9
E22	Volatility filescan (Win7SP1x86_23418)	grep -i "secret_file"; grep -i -E "secret|Downloads.*docx"	Both searches returned no output (negative result).	Fig. 30, Fig. 31
E23	Volatility shellbags --output=html (Win7SP1x86_23418)	IEUser UsrClass.dat: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0	C:\Users\IEUser\Downloads: key LastWrite 2019-01-06 15:04:40 UTC; Access field 2019-01-06 14:49:40 UTC; no F:\ entry; D:\ and E:\ entries last written in 2015; Desktop bag (ntuser.dat) LastWrite 2019-01-06 14:32:53 UTC.	Fig. 30, Fig. 31, Fig. 32 App. C12

Appendix C: Selected Volatility Output

C1 - E02 imageinfo

Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
AS Layer1 : IA32PagedMemoryPae (Kernel AS)
AS Layer2 : FileAddressSpace (/home/kali/DF204_CaseStudy3/memdump_working.mem)
PAE type : PAE
DTB : 0x185000L
KDBG : 0x82972c68L
Number of Processors : 1
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0x82973d00L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2019-01-06 15:09:06 UTC+0000
Image local date and time : 2019-01-06 07:09:06 -0800

C2 - E03 pslist (selected rows)

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start
0x8413a908	System	4	0	73	518	------	0	2019-01-06 18:02:44 UTC+0000
0x851ec488	smss.exe	236	4	2	29	------	0	2019-01-06 18:02:44 UTC+0000
0x84203998	sshd.exe	2016	1984	4	100	0	0	2019-01-06 15:02:49 UTC+0000
0x85bba860	explorer.exe	2524	2500	32	952	1	0	2019-01-06 15:02:54 UTC+0000
0x8583a030	chrome.exe	2388	2524	33	819	1	0	2019-01-06 15:04:30 UTC+0000
0x851e8520	chrome.exe	1280	2388	7	82	1	0	2019-01-06 15:04:30 UTC+0000
0x850316b8	chrome.exe	2380	2388	2	57	1	0	2019-01-06 15:04:30 UTC+0000
0x842b4d28	chrome.exe	2912	2388	9	159	1	0	2019-01-06 15:04:32 UTC+0000
0x8431fb08	cmd.exe	3996	2524	1	22	1	0	2019-01-06 15:06:11 UTC+0000
0x842ddd28	conhost.exe	3988	348	2	52	1	0	2019-01-06 15:06:11 UTC+0000
0x842eb6b0	chrome.exe	976	2388	14	198	1	0	2019-01-06 15:06:23 UTC+0000
0x84312030	FTK Imager.exe	2596	2524	14	358	1	0	2019-01-06 15:07:15 UTC+0000
0x850d5a38	dllhost.exe	2228	552	6	170...38	1	0	2019-01-06 15:09:07 UTC+0000

C3 - E04 netscan (selected records)

Offset(P)	Proto	Local Address	Foreign Address	State	Pid	Owner	Created
0x3e043b10	TCPv4	192.168.56.8:49177	192.168.56.5:80	CLOSED	-1		
0x3fcba008	TCPv4	192.168.56.8:49178	192.168.56.5:80	CLOSED	-1		
0x3e081f50	UDPv4	192.168.56.8:56647	*:*		2388	chrome.exe	2019-01-06 15:08:32 UTC+0000
0x3ee1bc18	TCPv4	0.0.0.0:22	0.0.0.0:0	LISTENING	2016	sshd.exe	

C4 - E05 TimeZoneInformation

Registry: \REGISTRY\MACHINE\SYSTEM
Key name: TimeZoneInformation (S)
Last updated: 2019-01-06 16:31:39 UTC+0000

REG_DWORD Bias : (S) 480
REG_SZ TimeZoneKeyName : (S) Pacific Standard Time
REG_DWORD ActiveTimeBias : (S) 480

C5 - E06 cmdscan (selected)

CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:

C6 - E07 consoles (selected)

ConsoleProcess: conhost.exe Pid: 3988
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
...
C:\Users\IEUser>ipconfig
IPv4 Address. . . . . : 192.168.56.8
...
C:\Users\IEUser>cd Downloads

C:\Users\IEUser\Downloads>copy secret_file.docx F:
1 file(s) copied.

C:\Users\IEUser\Downloads>

C7 - E09 getsids -p 3996

cmd.exe (3996): S-1-5-21-1716914095-909560446-1177810406-1000 (IEUser)
cmd.exe (3996): S-1-5-21-1716914095-909560446-1177810406-513 (Domain Users)
cmd.exe (3996): S-1-1-0 (Everyone)
cmd.exe (3996): S-1-5-114 (Local Account (Member of Administrators))
cmd.exe (3996): S-1-5-32-544 (Administrators)
cmd.exe (3996): S-1-5-32-545 (Users)
cmd.exe (3996): S-1-5-4 (Interactive)
cmd.exe (3996): S-1-2-1 (Console Logon (Users who are logged onto the physical console))
cmd.exe (3996): S-1-5-11 (Authenticated Users)
cmd.exe (3996): S-1-5-15 (This Organization)
cmd.exe (3996): S-1-5-113 (Local Account)
cmd.exe (3996): S-1-5-5-0-61708 (Logon Session)

C8 - E08 hivelist (selected rows)

Virtual Physical Name
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260
\??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat

C9 - E10, E11, E12, E18-E21 USB and drive-mapping registry keys (selected)

[E10] SYSTEM ControlSet001\Enum\USBSTOR Last updated: 2019-01-06 15:02:54 UTC+0000
Subkeys: (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

[E20] SYSTEM ...\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0 Last updated: 2019-01-06 15:02:54 UTC+0000
Subkeys: Device Parameters, LogConf, Properties, Control
REG_SZ ContainerID : (S) {421e7e37-11dd-11e9-b3cf-806e6f6e6963}
REG_SZ FriendlyName : (S) General UDisk USB Device
REG_MULTI_SZ HardwareID : (S) ['USBSTOR\\DiskGeneral_UDisk_____5.00', ...]

[E21] SYSTEM ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{...}
Last updated: 2019-01-06 15:02:54 UTC+0000
REG_SZ DeviceInstance : (S) USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0

[E18] SYSTEM ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b} Last updated: 2019-01-06 15:02:54 UTC+0000

[E11] SYSTEM MountedDevices Last updated: 2019-01-06 15:02:54 UTC+0000
\DosDevices\F: (hex dump, decoded as UTF-16LE by the examiner):
_??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
\??\Volume{421e7e52-11dd-11e9-b3cf-08002709e15d}: same USBSTOR string
\DosDevices\E: VirtualBox CD-ROM (IDE#CdRomVBOX_CD-ROM...)

[E12] \??\C:\Users\IEUser\ntuser.dat Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 Last updated: 2019-01-06 15:03:07 UTC+0000
Subkeys: CPC, F, {421e7e52-11dd-11e9-b3cf-08002709e15d}, {762f4ebc-60ea-11e5-83af-806e6f6e6963},
{8358fe6d-60fa-11e5-bb4a-806e6f6e6963}, {a5b8a980-608c-11e5-a266-806e6f6e6963}, {a5b8a983-608c-11e5-a266-806e6f6e6963}

[E19] ...\MountPoints2\F Last updated: 2019-01-06 15:03:06 UTC+0000
...\MountPoints2\{421e7e52-11dd-11e9-b3cf-08002709e15d} Last updated: 2019-01-06 15:03:07 UTC+0000 Subkeys: (S) shell

C10 - E13 / E16 strings mapped to processes (selected; long values shortened with ...)

152263320 [2388:072eda98] http://192.168.56.5/secret_file.docx
568756597 [2388:0720c975] http://192.128.56.5/http://192.168.56.5/secret_file.docx"2d0e-57ea40b47d94c"Fri, 04 Jan 2019 16:05:12 GMTapplication/vnd.openxmlformats-officedocument.wordprocessingml.document...
799213234 [2388:060bc6b2] tps://www.google.com/complete/search?client=chrome-omni&...&q=192.168.56.5%2Fsecret_file.docx&oit=3&...
353107270 [FREE MEMORY:-1]
/secret_file.docxhttps://www.google.com/search?q=192.168.56.5%2Fsecret_file.docx&oq=...
762519243 [1228:02331ecb 2388:010acecb] http://192.168.56.5/secret_file.docx)

C11 - E14 / E15 / E17 accounts

[E14] SOFTWARE Microsoft\Windows NT\CurrentVersion\ProfileList Last updated: 2015-09-21 09:50:52 UTC+0000
Subkeys: S-1-5-18, S-1-5-19, S-1-5-20, S-1-5-21-1716914095-909560446-1177810406-1000, S-1-5-21-1716914095-909560446-1177810406-1002
[E17] ...-1000 ProfileImagePath: C:\Users\IEUser Last updated: 2019-01-06 15:07:15 UTC+0000
[E17] ...-1002 ProfileImagePath: C:\Users\sshd_server Last updated: 2015-09-21 09:51:49 UTC+0000
[E15] SAM SAM\Domains\Account\Users\Names Last updated: 2015-09-21 09:50:50 UTC+0000
Subkeys: Administrator, Guest, IEUser, sshd, sshd_server

C12 - E23 ShellBags (selected records from shellbags.html)

Registry : \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key : Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0
LastWrite : 2019-01-06 15:04:40 UTC+0000
FileName : DOWNLO~1 Create: 2015-09-21 09:17:34 UTC+0000 Access: 2019-01-06 14:49:40 UTC+0000
Attributes: RO, DIR Path : C:\Users\IEUser\Downloads

Other removable-looking paths in the output: D:\32Bit, D:\64Bit (BagMRU\0\3, LastWrite 2015-09-22 08:08:43) and E:\support (BagMRU\0\2, LastWrite 2015-09-21 22:30:28).
No F:\ entry appears in the output.
Desktop bag (ntuser.dat, Software\Microsoft\Windows\Shell\Bags\1\Desktop) LastWrite: 2019-01-06 14:32:53 UTC+0000